

Printer Dover

Spruce version 11.2 -- spooler version 11.2

File: DispY-Rev-Cj.sil etc.

Creation date: 22-Oct-81 10:23:28

For: Spruce

33 total sheets = 32 pages, 1 copy.

Problems encountered:

Font HELVETICA24B substituted for font HELVE24.

Not impl.: brightness, hue, saturation, show-object, show-dots-opaque, dots from files.

Not impl.: brightness, hue, saturation, show-object, show-dots-opaque, dots from files.

Not impl.: brightness, hue, saturation, show-object, show-dots-opaque, dots from files.

Not impl.: brightness, hue, saturation, show-object, show-dots-opaque, dots from files.

Not impl.: brightness, hue, saturation, show-object, show-dots-opaque, dots from files.

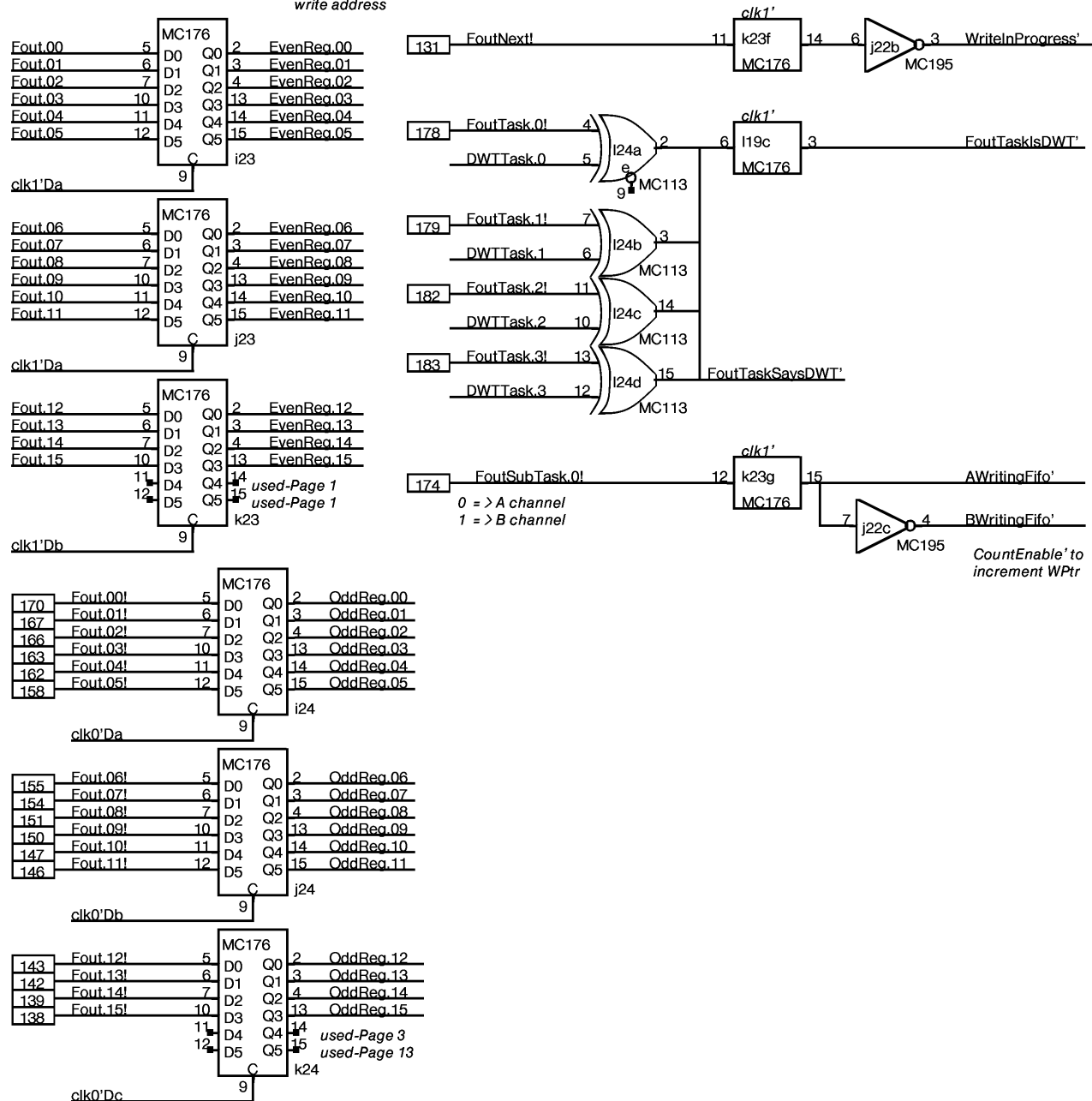
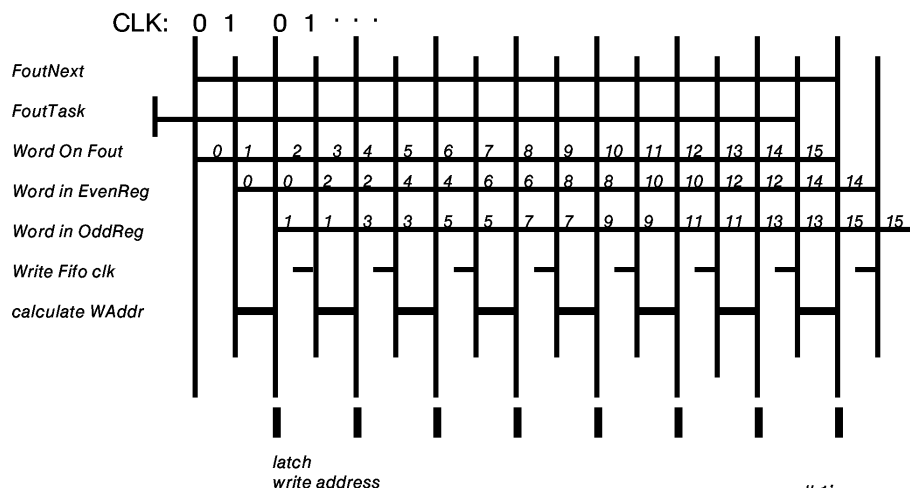
Not impl.: brightness, hue, saturation, show-object, show-dots-opaque, dots from files.

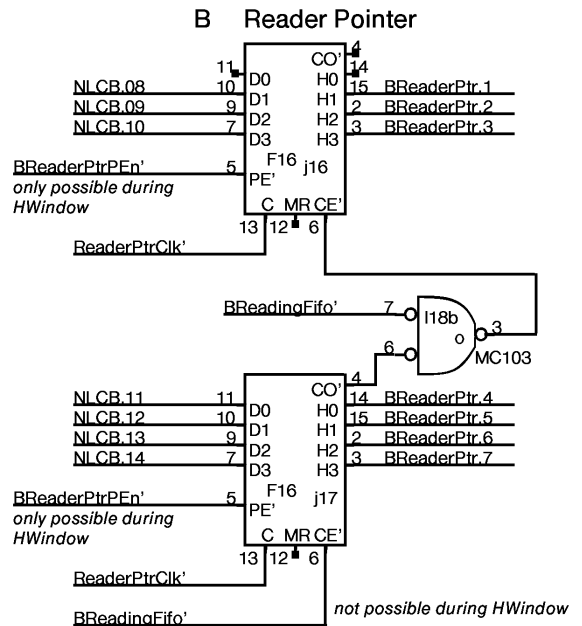
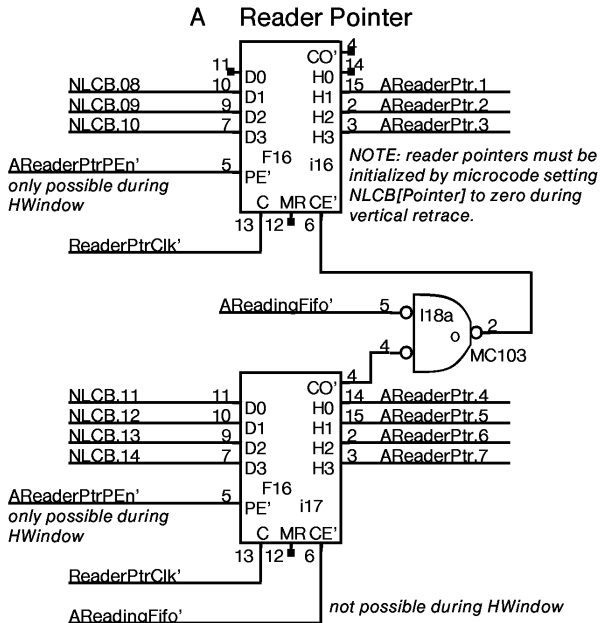
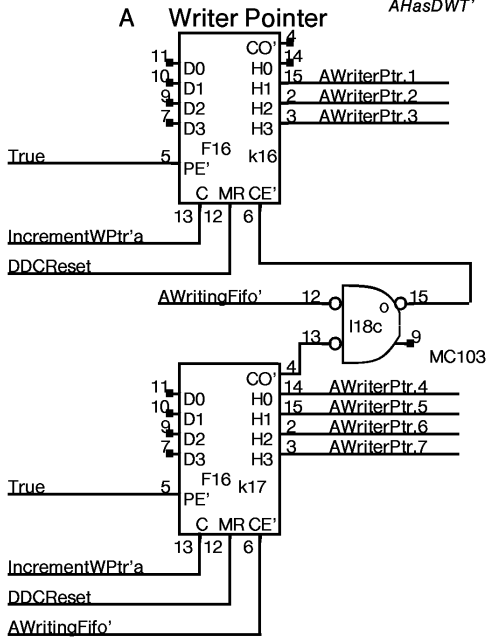
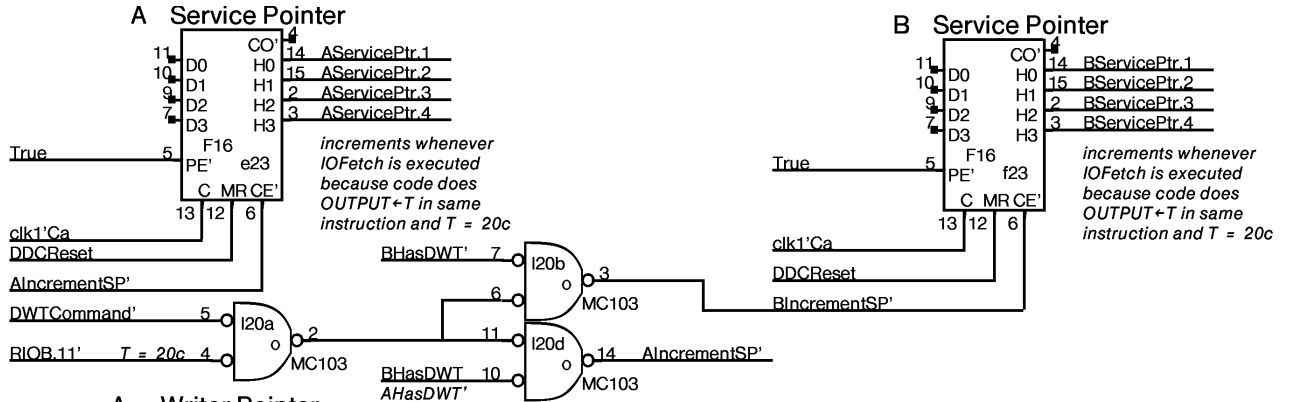
DORADO SCHEMATICS

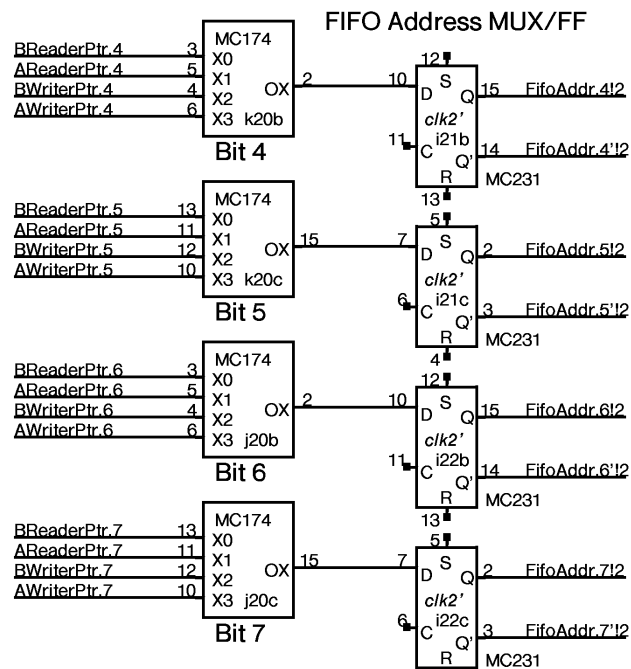
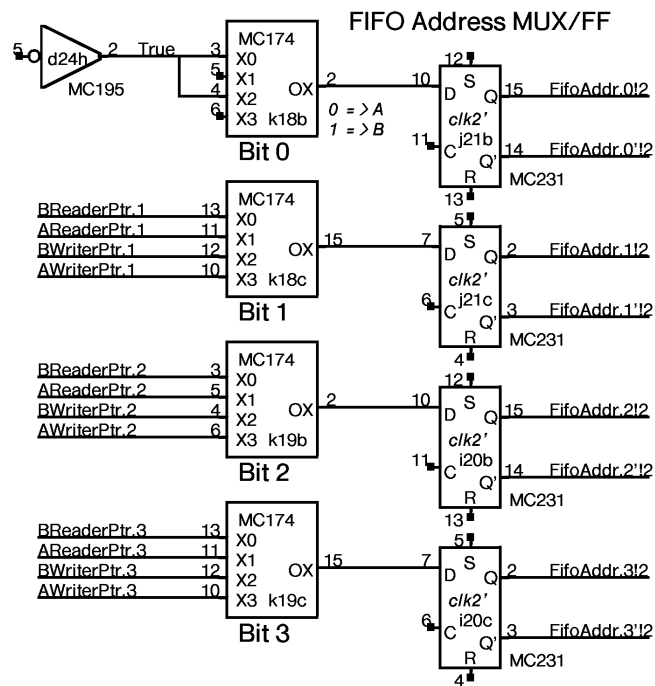
Display Y

Table of contents

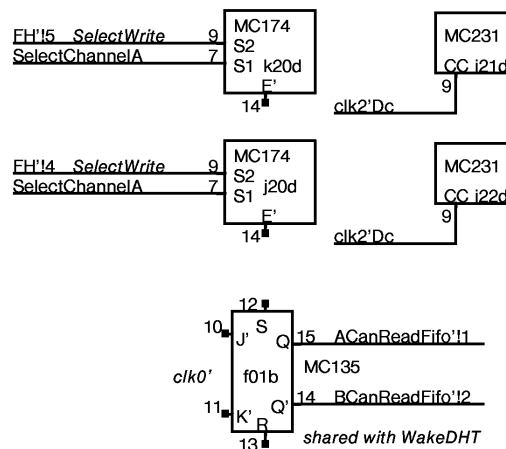
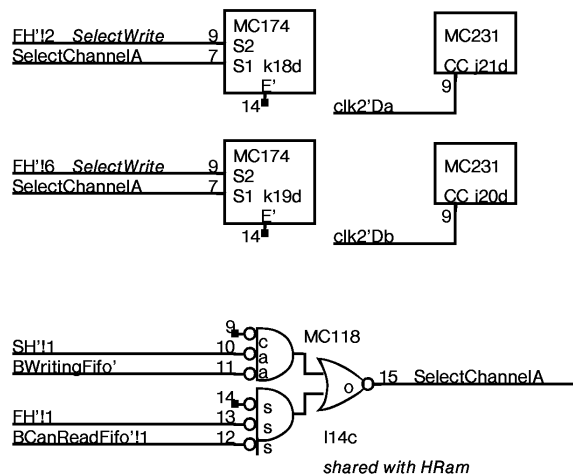
<u>TITLE</u>	<u>Page</u>
FOUT interface _____	01
Pointers _____	02
Fifo Address Logic _____	03
Fifo _____	04
Item Permutation Logic _____	05
Intermediate Buffer Control _____	06
Channel A Data Path _____	07
Channel A Items and Shift/Load Control _____	08
Channel A Current Line Control Block _____	09
Channel B Data Path _____	10
Channel B Items and Shift/Load Control _____	11
Channel B Current Line Control Block _____	12
Next Line Control Block _____	13
Horizontal RAM _____	14
Flags _____	15
DWT Task WakeUp Logic _____	16
Cursor _____	17
Slow IO Interface _____	18
Alto Display Drivers _____	19
MiniMixer _____	20
OIS Terminal Interface _____	21
Statics Register and Muffler Control _____	22
Mufflers _____	23
Clock Drivers _____	24
Pre Clock Drivers _____	25
Layout _____	26
Slow IO System Formats _____	27
Next Line Control Block Formats _____	28
DDC to DDM Interface Table _____	29
Cabling Summary _____	30
Configuration _____	31







Write In FH
Read In SH

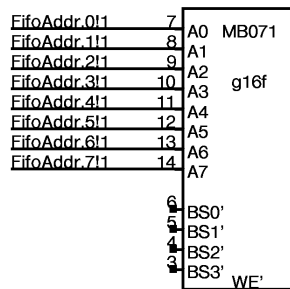


EvenReg.00	19	g16b	23	Fifo.00
		MB071		
EvenReg.01	18	g16c	22	Fifo.01
		MB071		
EvenReg.02	17	g16d	21	Fifo.02
		MB071		
EvenReg.03	16	g16e	20	Fifo.03
		MB071		
EvenReg.04	19	g18b	23	Fifo.04
		MB071		
EvenReg.05	18	g18c	22	Fifo.05
		MB071		
EvenReg.06	17	g18d	21	Fifo.06
		MB071		
EvenReg.07	16	g18e	20	Fifo.07
		MB071		
EvenReg.08	19	g20b	23	Fifo.08
		MB071		
EvenReg.09	18	g20c	22	Fifo.09
		MB071		
EvenReg.10	17	g20d	21	Fifo.10
		MB071		
EvenReg.11	16	g20e	20	Fifo.11
		MB071		
EvenReg.12	19	g22b	23	Fifo.12
		MB071		
EvenReg.13	18	g22c	22	Fifo.13
		MB071		
EvenReg.14	17	g22d	21	Fifo.14
		MB071		
EvenReg.15	16	g22e	20	Fifo.15
		MB071		

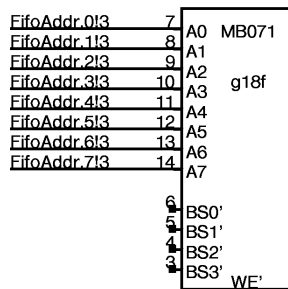
EVEN WORDS

OddReg.00	19	h16b	23	Fifo.16
		MB071		
OddReg.01	18	h16c	22	Fifo.17
		MB071		
OddReg.02	17	h16d	21	Fifo.18
		MB071		
OddReg.03	16	h16e	20	Fifo.19
		MB071		
OddReg.04	19	h18b	23	Fifo.20
		MB071		
OddReg.05	18	h18c	22	Fifo.21
		MB071		
OddReg.06	17	h18d	21	Fifo.22
		MB071		
OddReg.07	16	h18e	20	Fifo.23
		MB071		
OddReg.08	19	h20b	23	Fifo.24
		MB071		
OddReg.09	18	h20c	22	Fifo.25
		MB071		
OddReg.10	17	h20d	21	Fifo.26
		MB071		
OddReg.11	16	h20e	20	Fifo.27
		MB071		
OddReg.12	19	h22b	23	Fifo.28
		MB071		
OddReg.13	18	h22c	22	Fifo.29
		MB071		
OddReg.14	17	h22d	21	Fifo.30
		MB071		
OddReg.15	16	h22e	20	Fifo.31
		MB071		

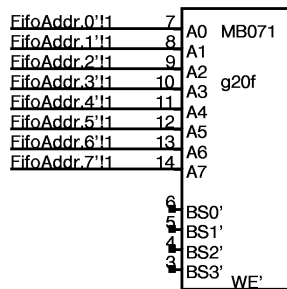
ODD WORDS



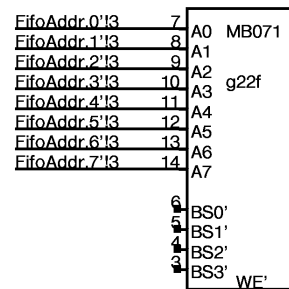
WriteFifo'a 2



WriteFifo'b 2

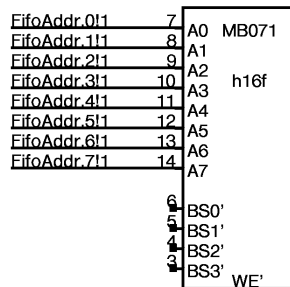


WriteFifo'c 2

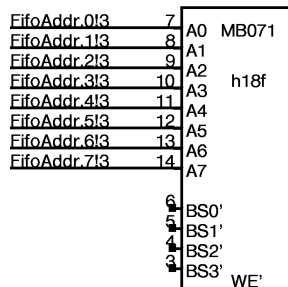


WriteFifo'd 2

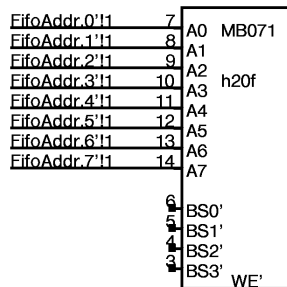
EVEN WORDS



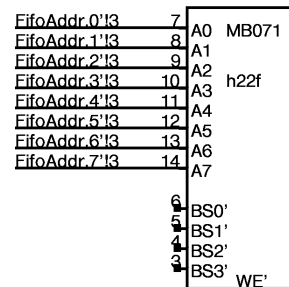
WriteFifo'a 2



WriteFifo'b 2



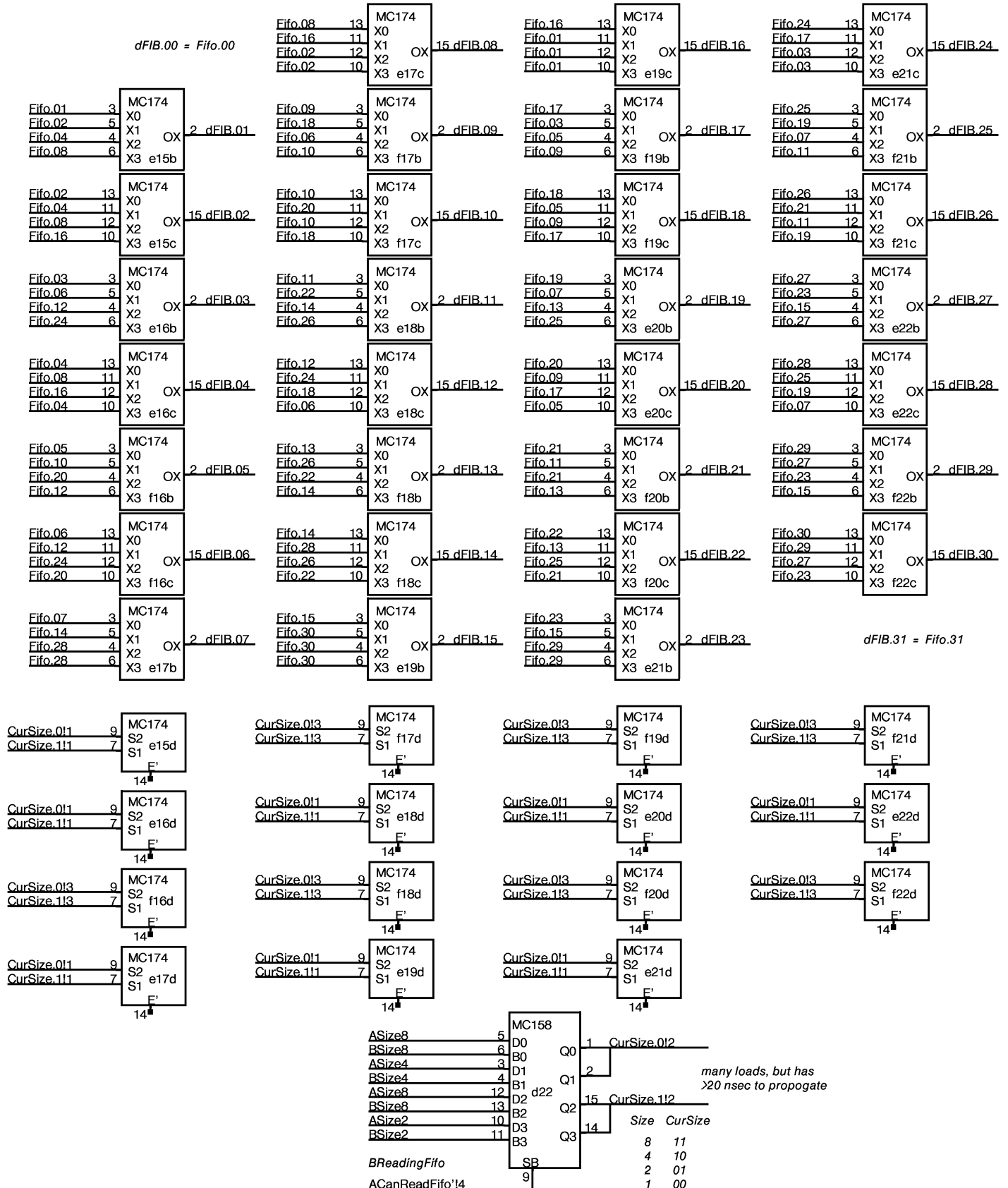
WriteFifo'c 2

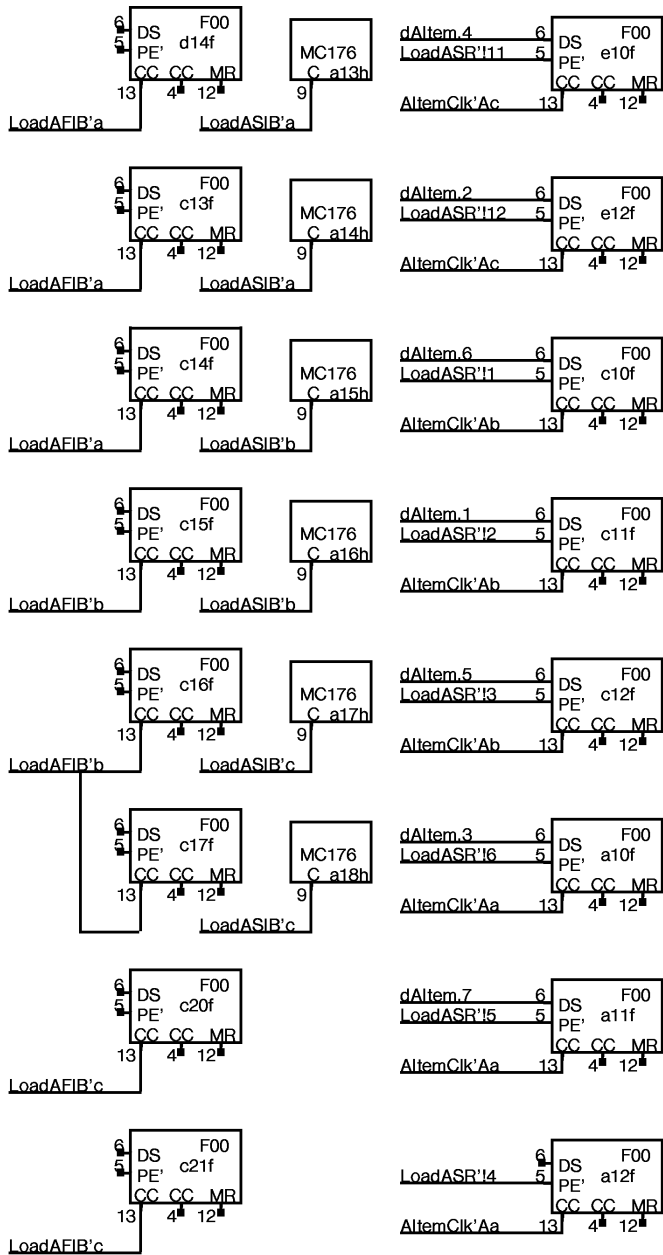


WriteFifo'd 2

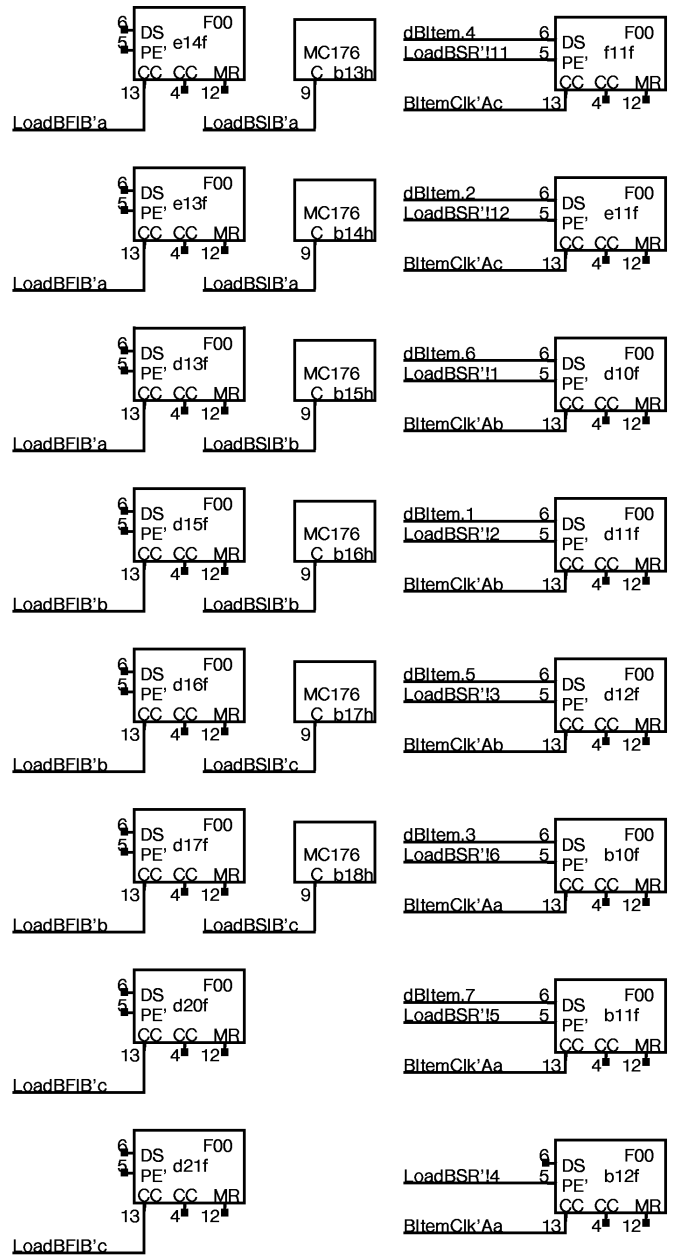
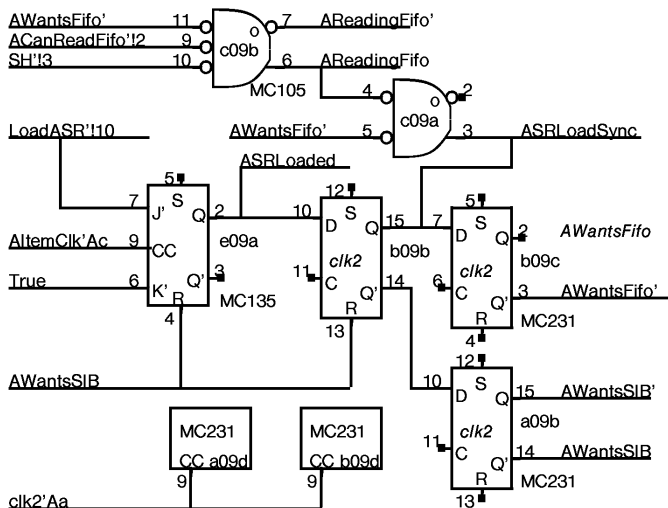
ODD WORDS

Item Generator Permuter
See ItemGenerator table for input specs

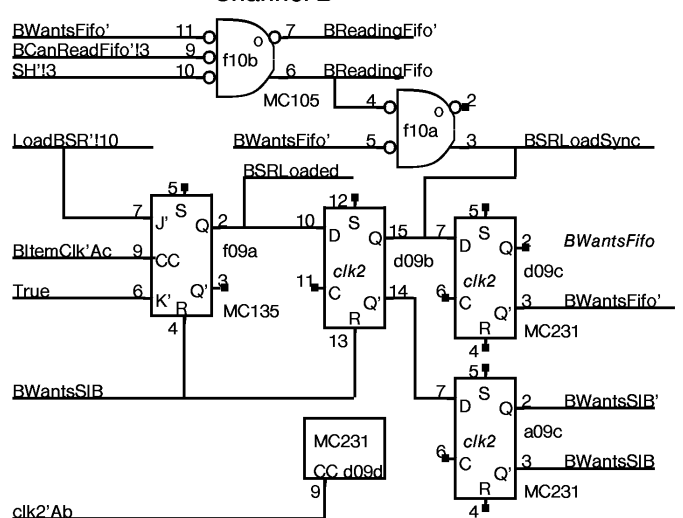


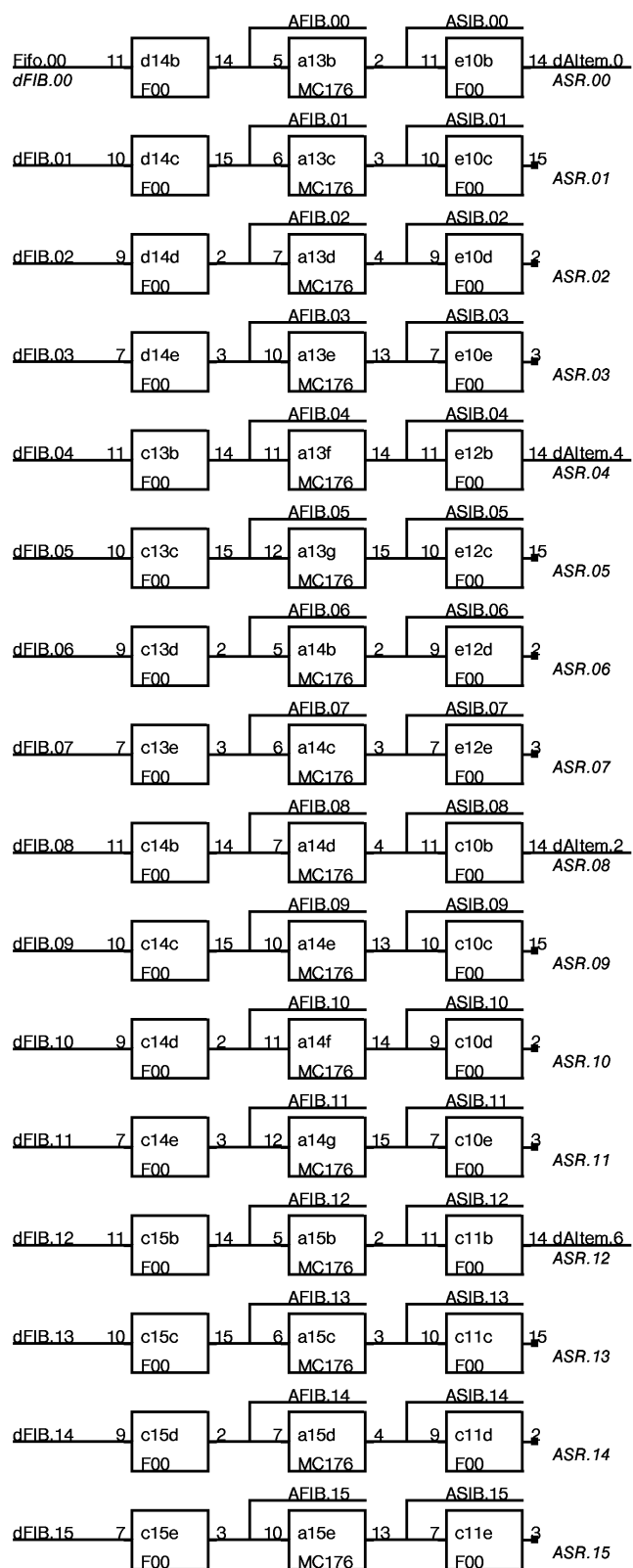


Channel A



Channel B

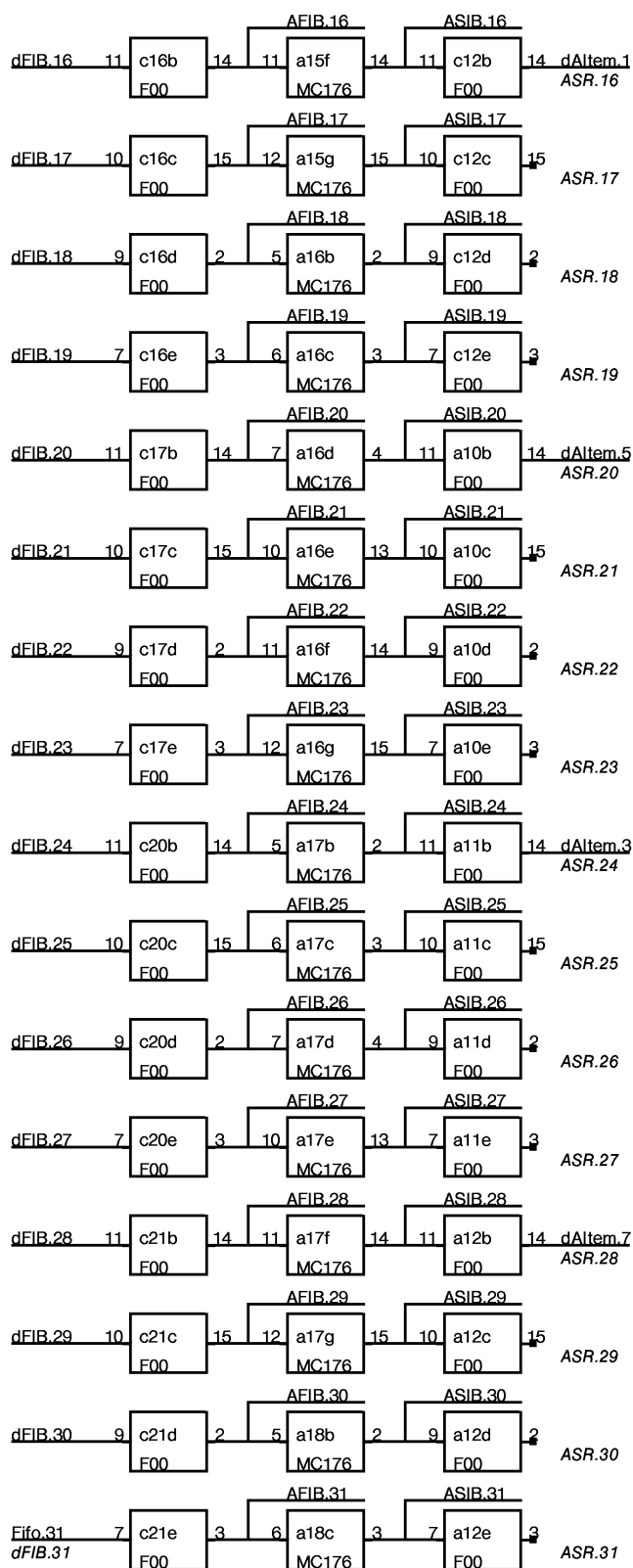




AFIB

ASIB

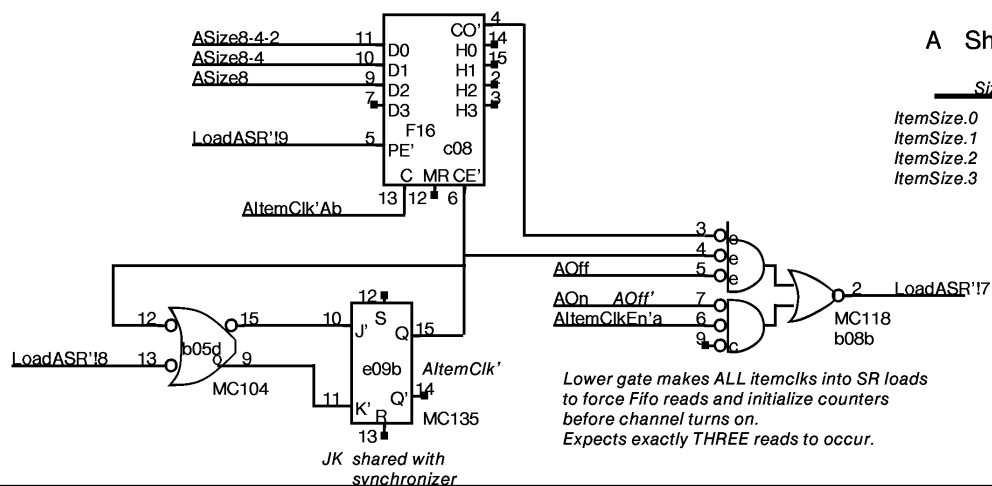
ASR



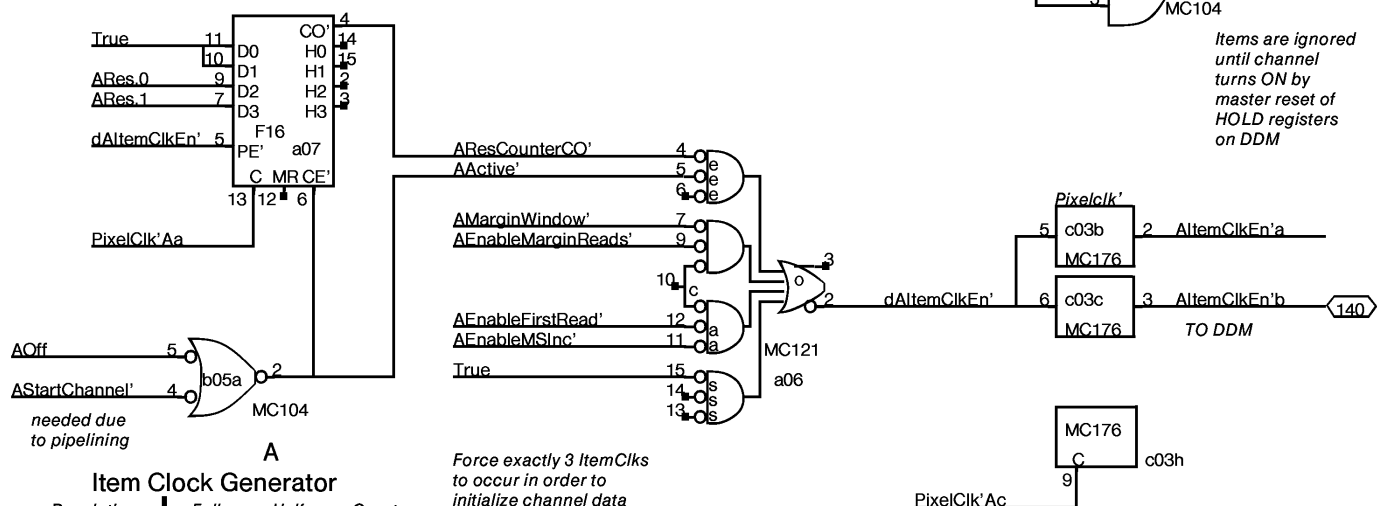
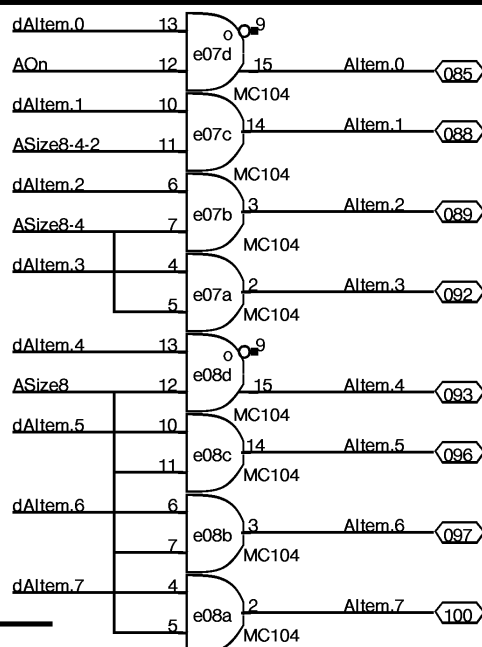
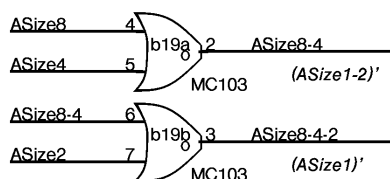
AFIB

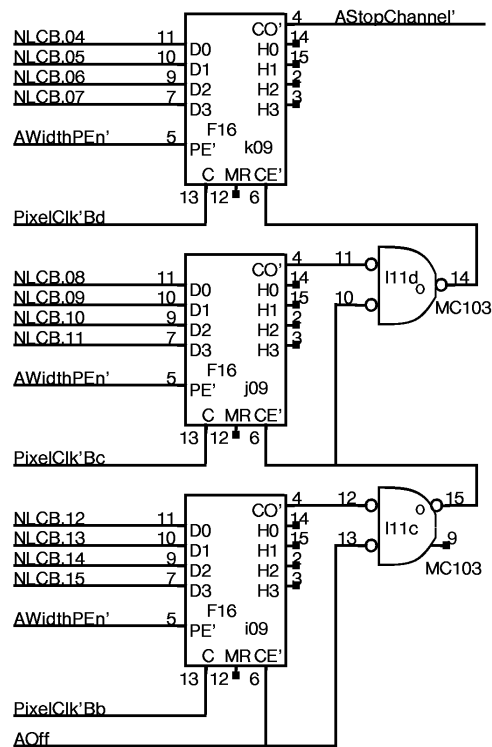
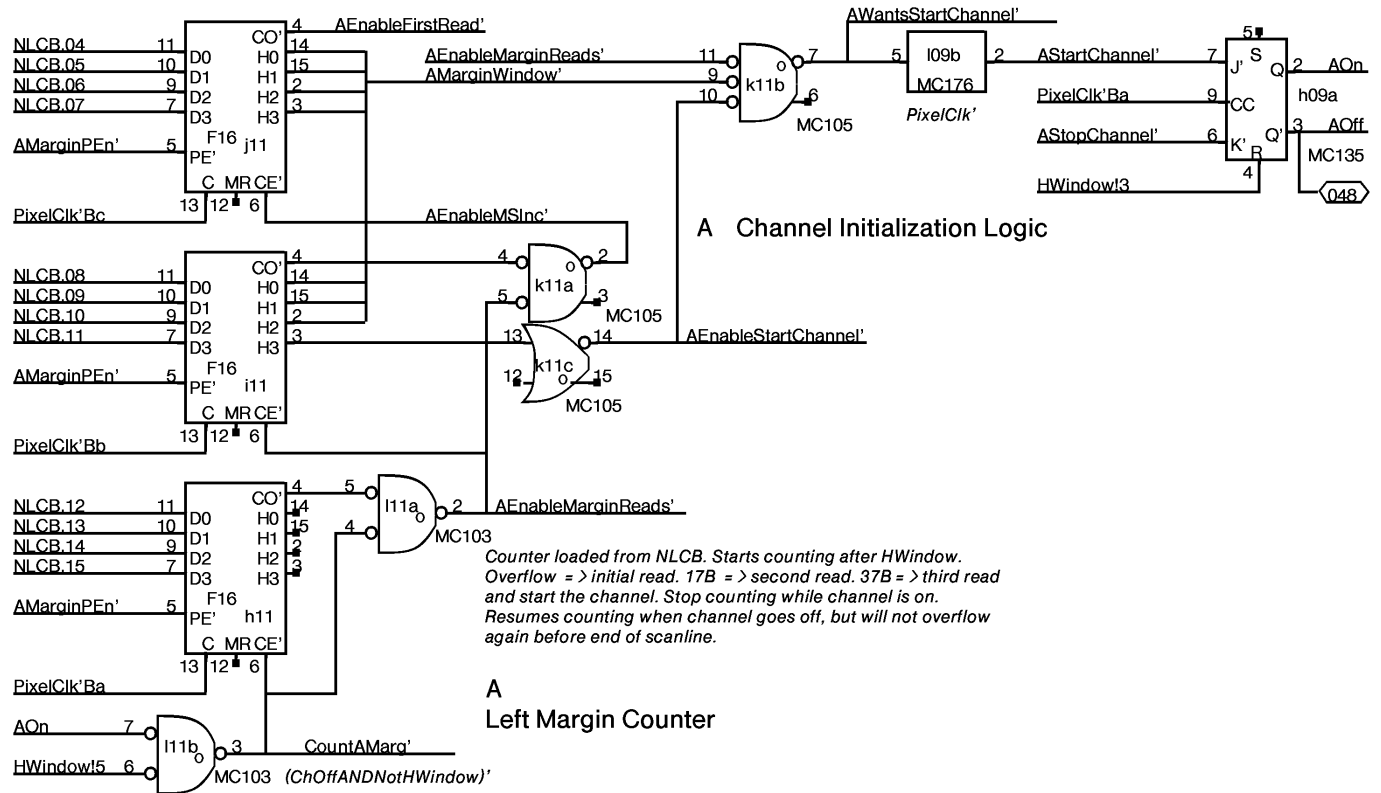
ASIB

ASR

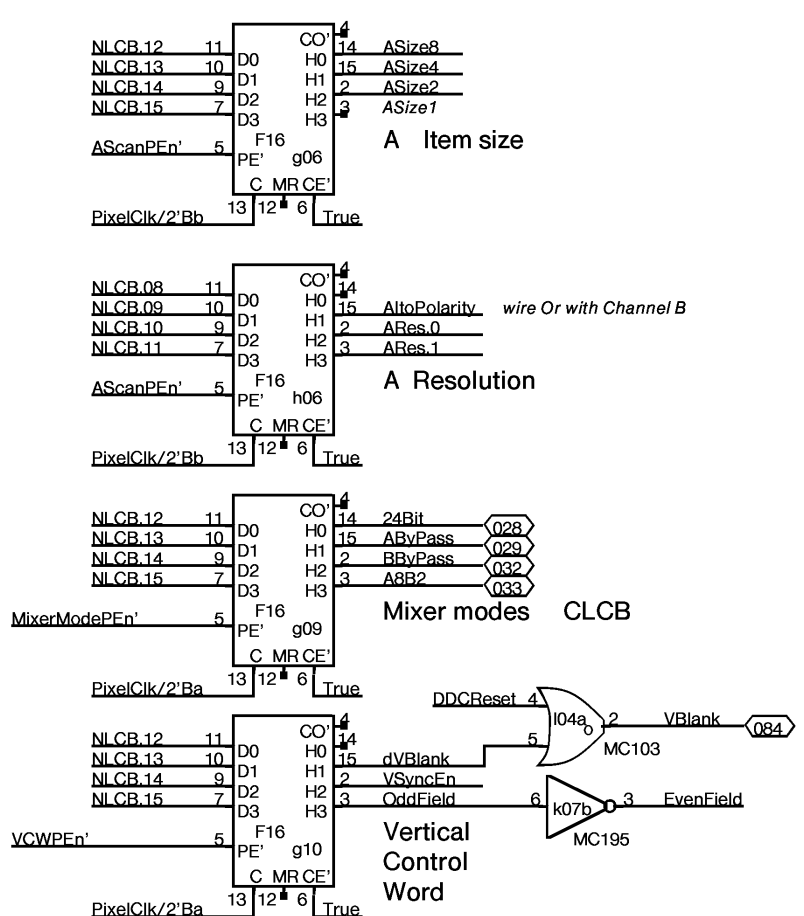


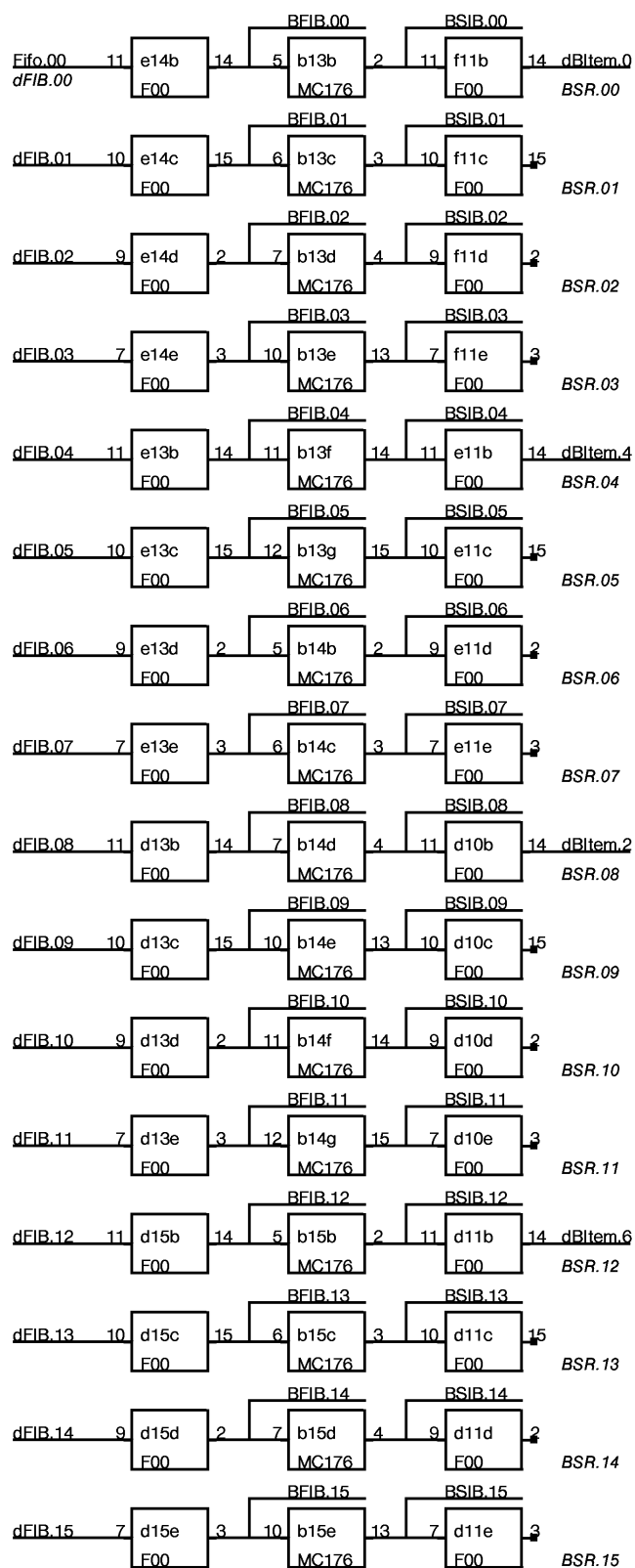
A Item buffer logic





Stop channel when counter value is > Z400B

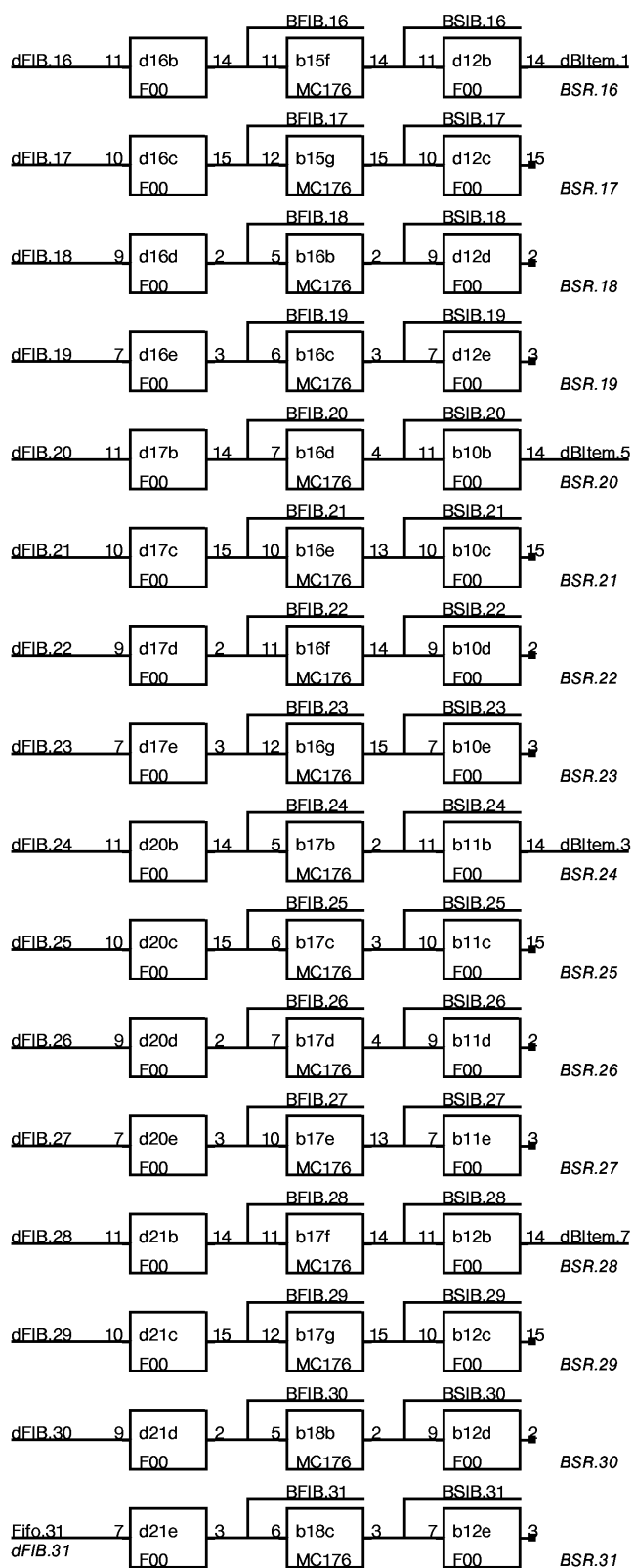




BFIB

BSIB

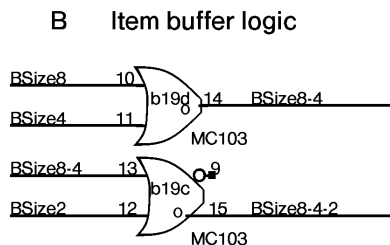
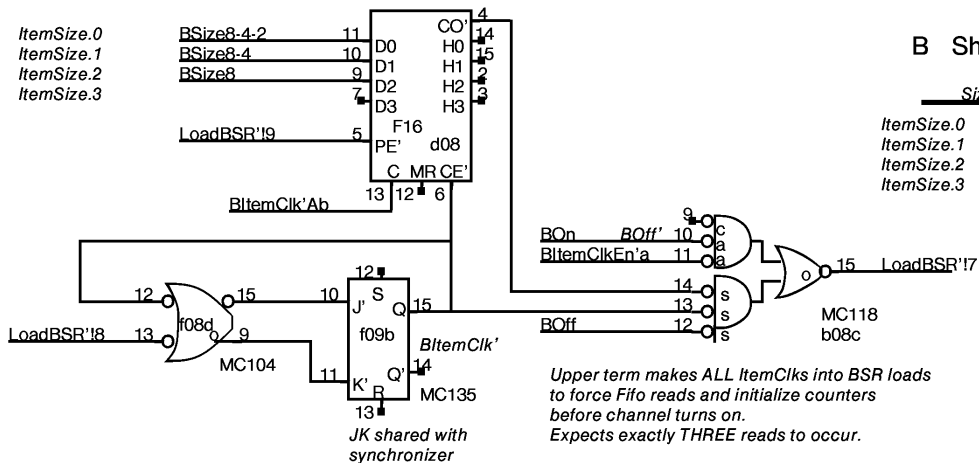
BSR



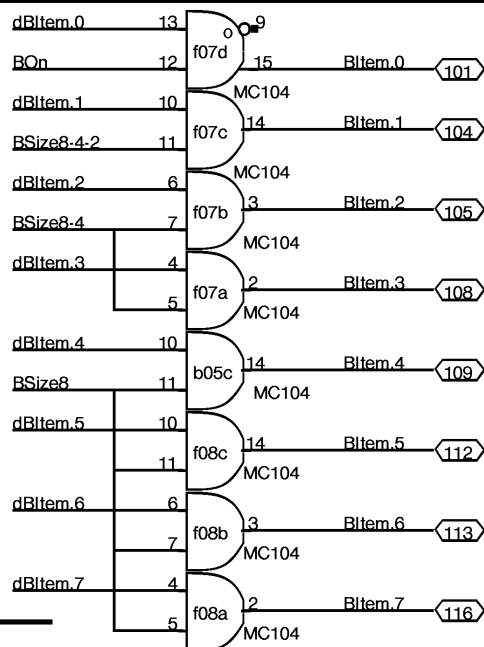
BFIB

BSIB

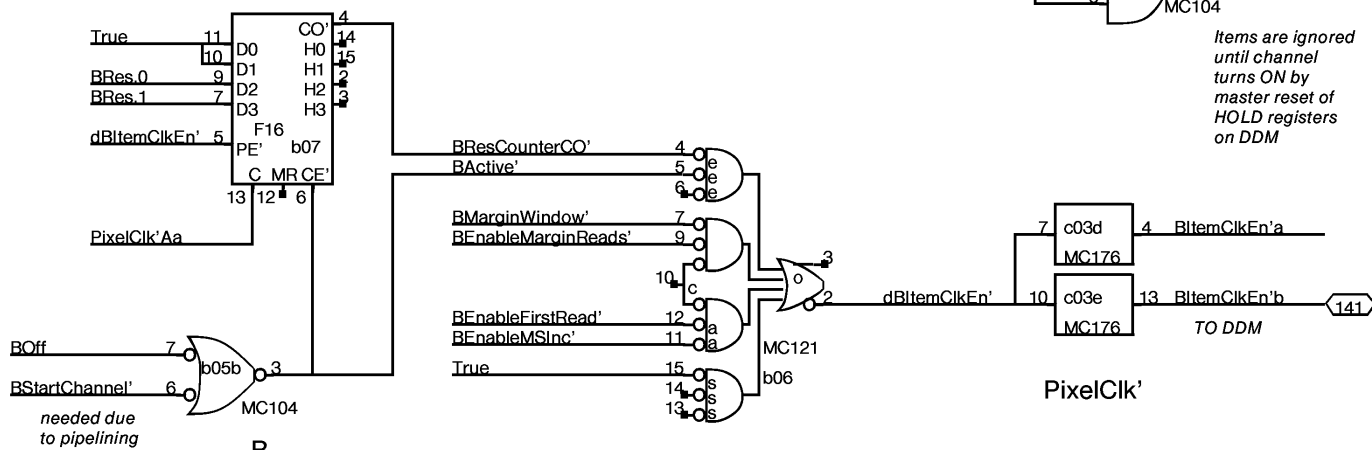
BSR



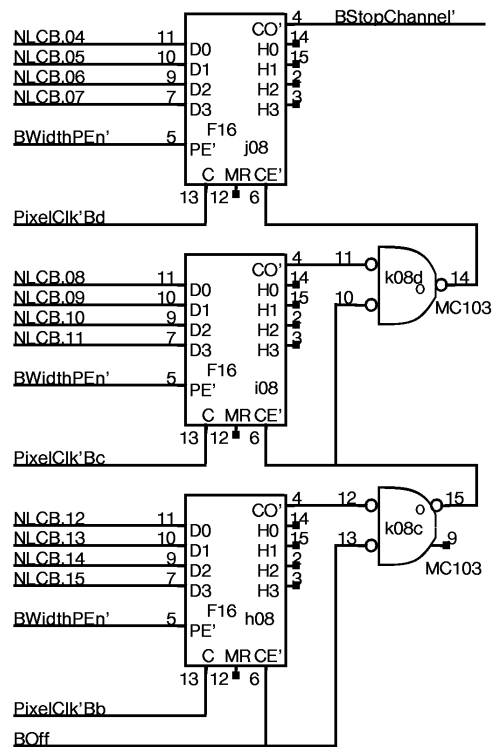
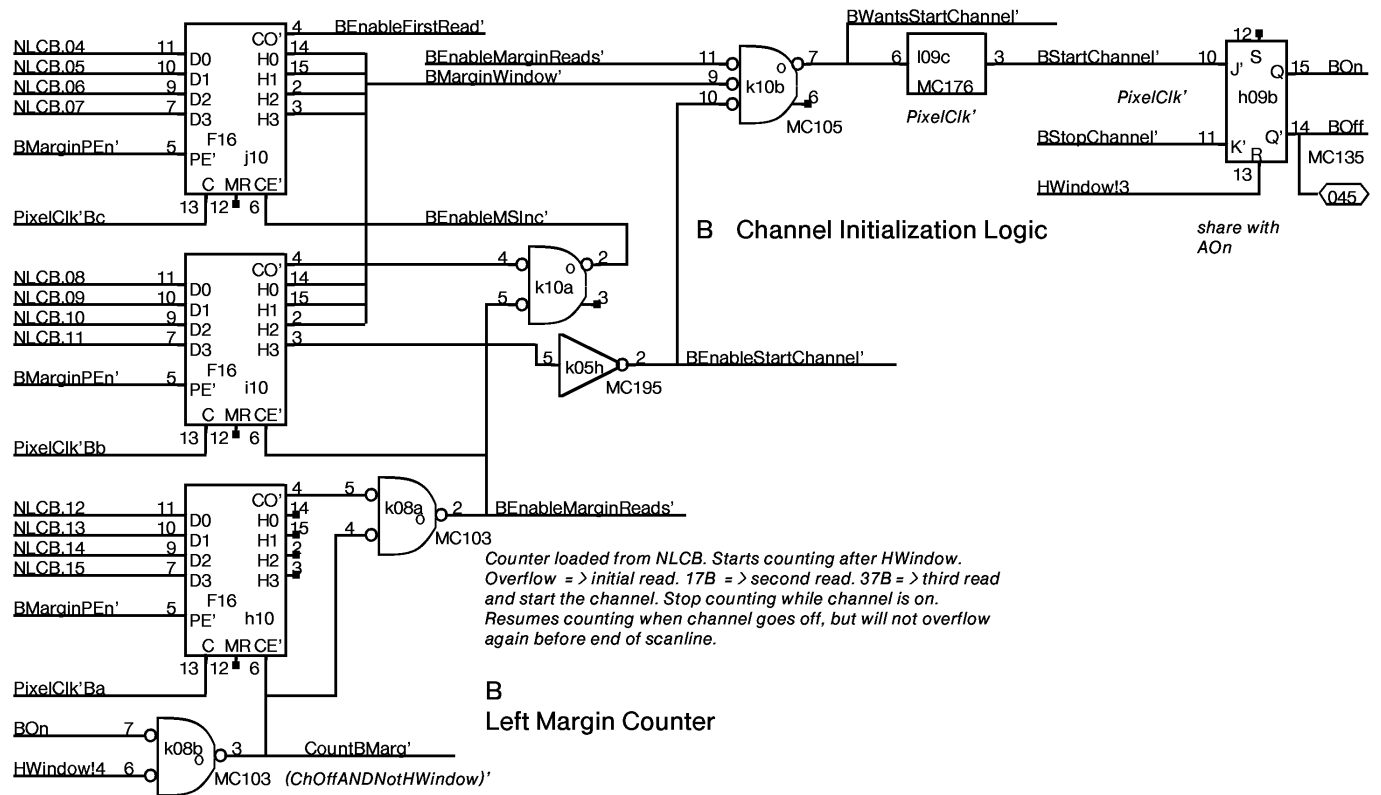
dItems are garbage until channel becomes active



Items are ignored until channel turns ON by master reset of HOLD registers on DDM

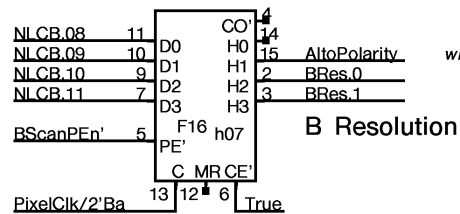
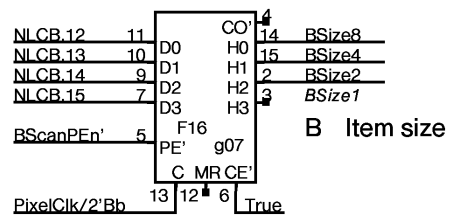


PixelClk'

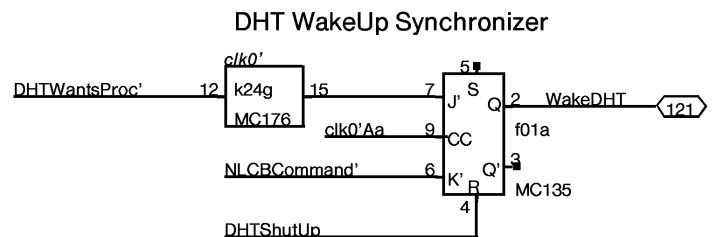
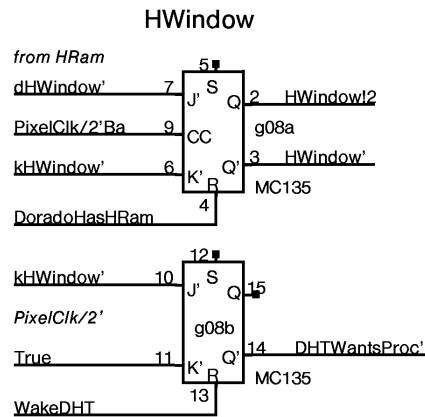
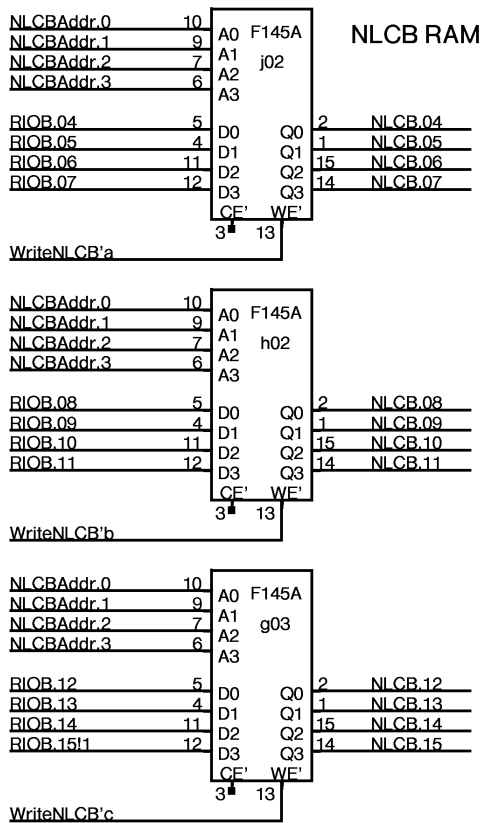
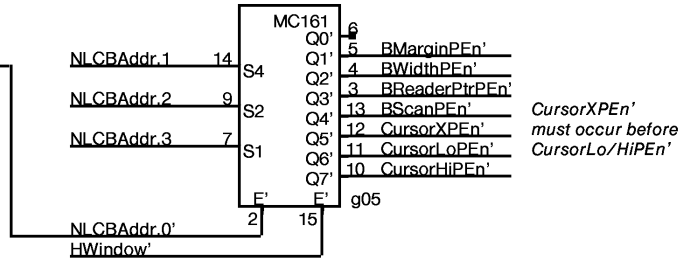
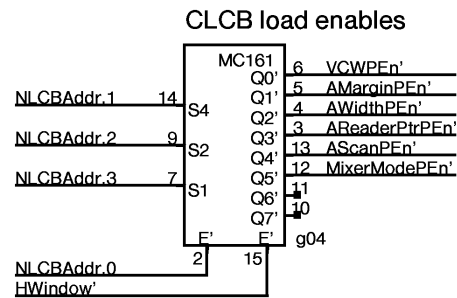
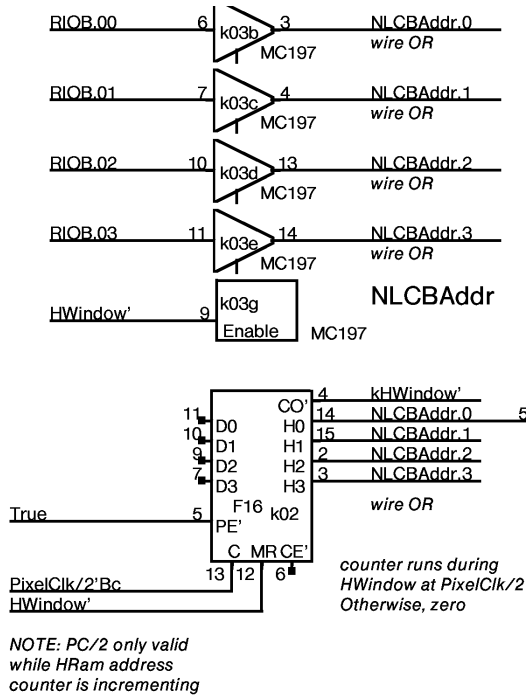


B
DCB Width Counter

Stop channel when
counter > Z400b

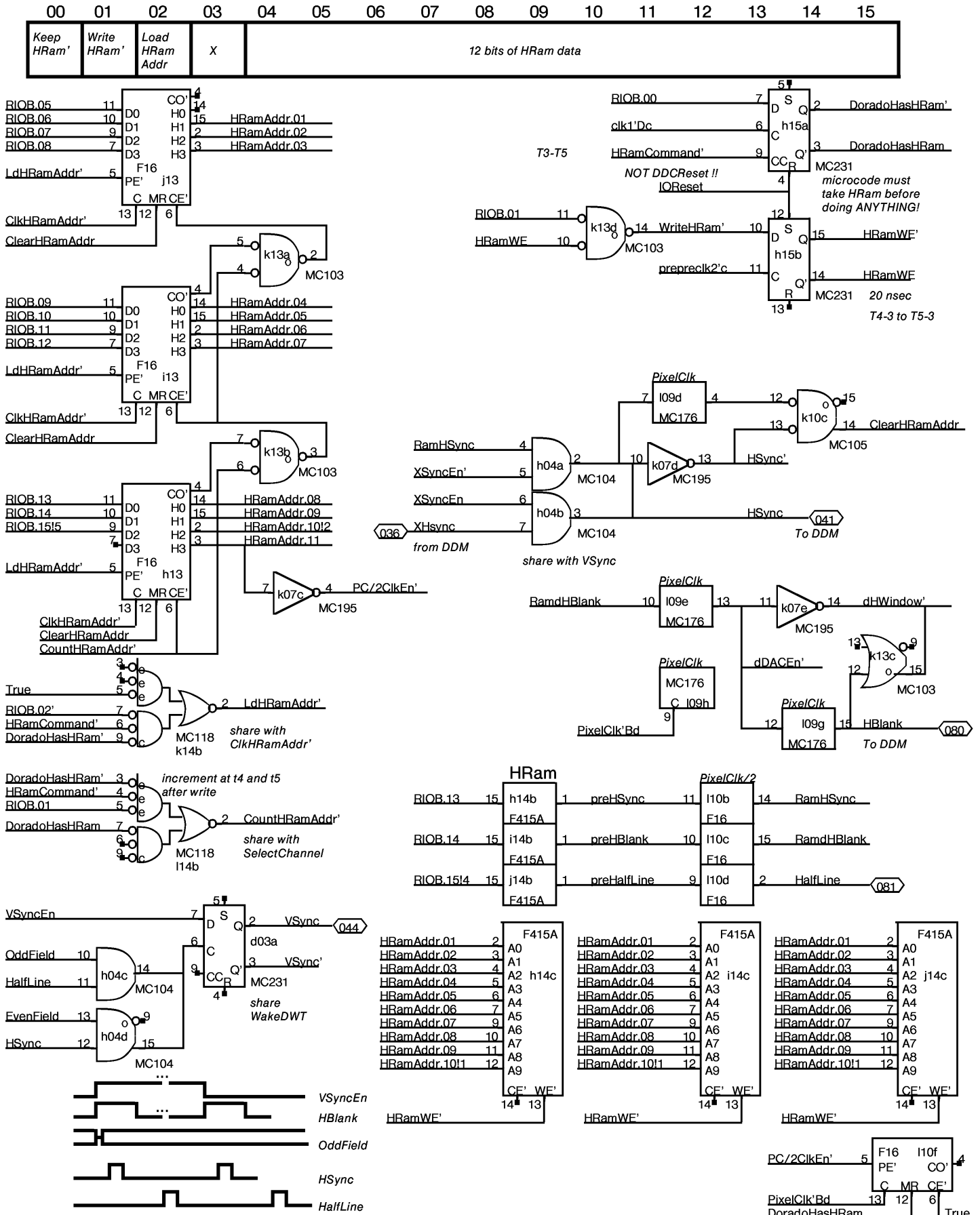


wire OR with Channel A



For convenience, use NLCB command to kill wakeup. Assumes DHT will always do some NLCB command whenever it is awakened. Default would be to load an unused, don't care address in NLCB.

requires minimum 4 instruction loop



THE DDC MAINTAINS, FOR EACH CHANNEL, TWO FLAGS:

NEXT Word Control Block Flag (NextWCBFlag)
Current Word Control Block Flag (CurrentWCBFlag)

A Word Control Block is a pair of values called Address and MunchCount, for either the CURRENT or the NEXT scanline.

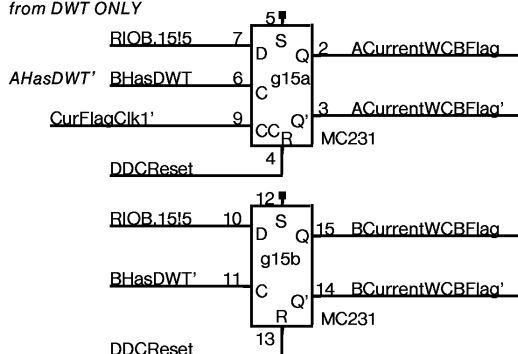
Flag management	SET	CLEARED
NextWCBFlag	by DHT when it has filled the NextWCB	by DWT when it has copied NextWCB into CurrentWCB
CurrentWCBFlag	by DWT when it has copied NextWCB into CurrentWCB	by DWT after it has sent out all the data from the CurrentWCB

WakeUp conditions:

WakeDHT: whenever CLCB ← NLCB i.e.: end of every HWindow

WakeDWT: (CurrentWCBFlag AND BufferAvailable) OR (NextWCBFlag AND NOT(CurrentWCBFlag))

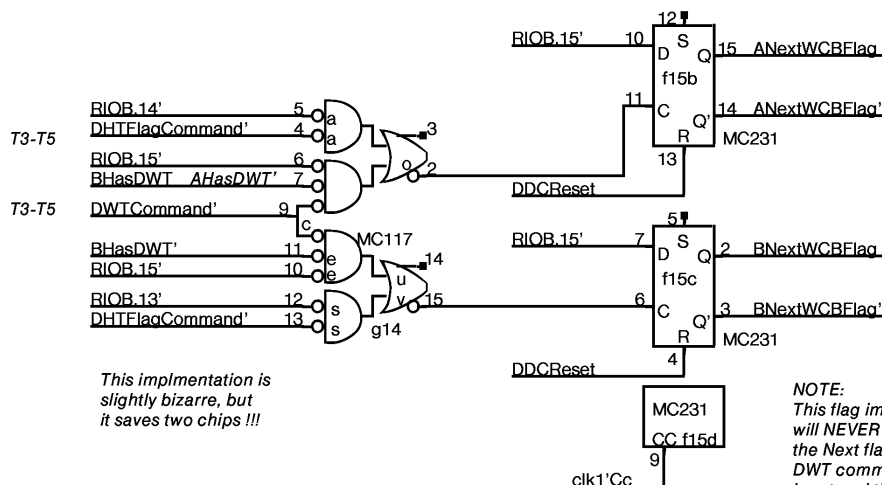
1 => Set CWCBFlag
0 => Clear CWCBFlag
from DWT ONLY



This circuit ASSUMES:
1. a DWT command with RI0B.11 set is to be ignored.
2. Any DWT command with RI0B.11 low is a CWCBFlag command, either set or clear.

See clock page.

See clock page.



This implementation is slightly bizarre, but it saves two chips !!!

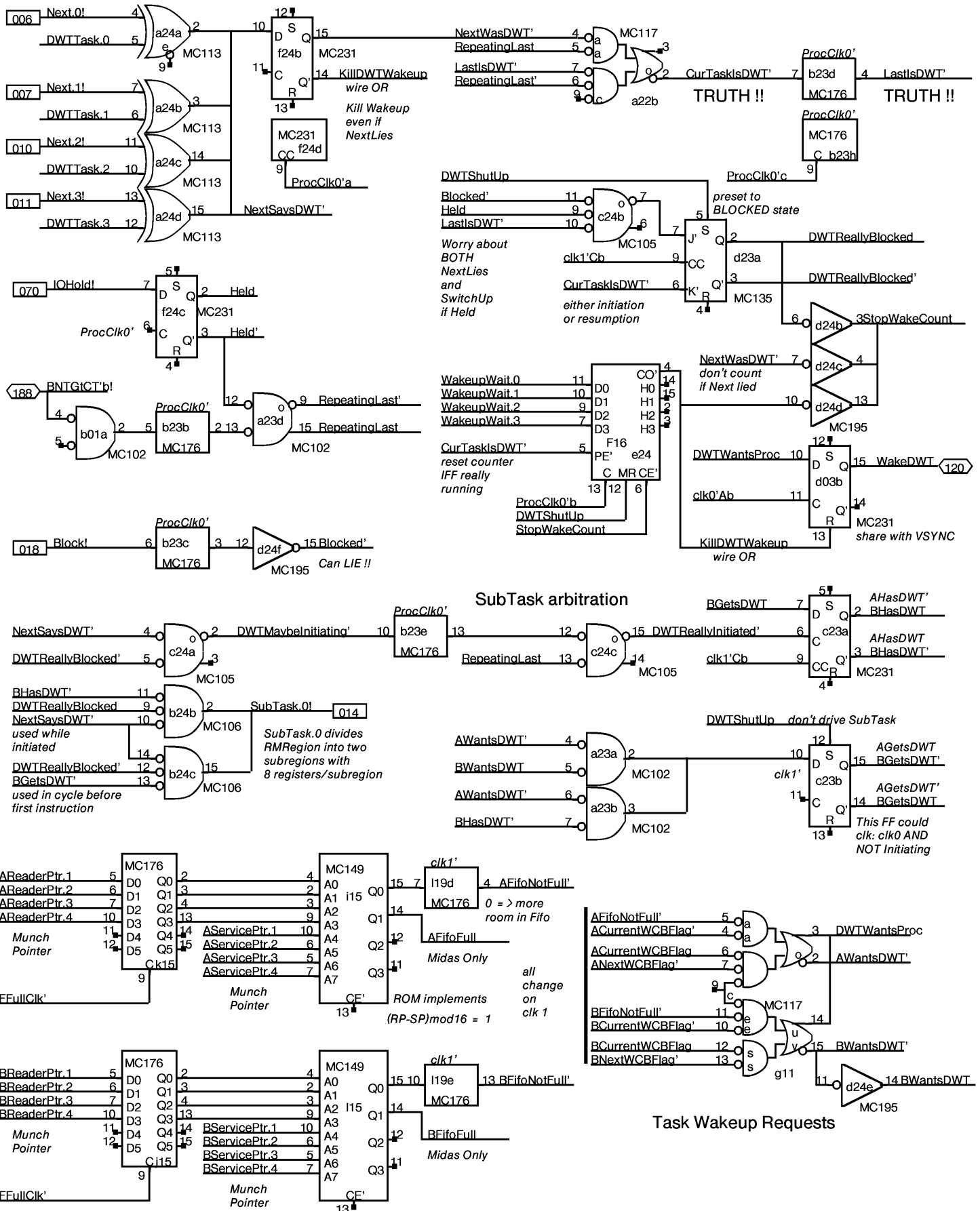
NOTE:
This flag implementation assumes that DHT flag commands will NEVER set bit 15 and will ALWAYS set either 13 or 14, the Next flag to set. Also, it assumes the ONLY kind of DWT command it should respond to is one in which bit 15 is set and therefore NextWCBFlag is to be cleared.

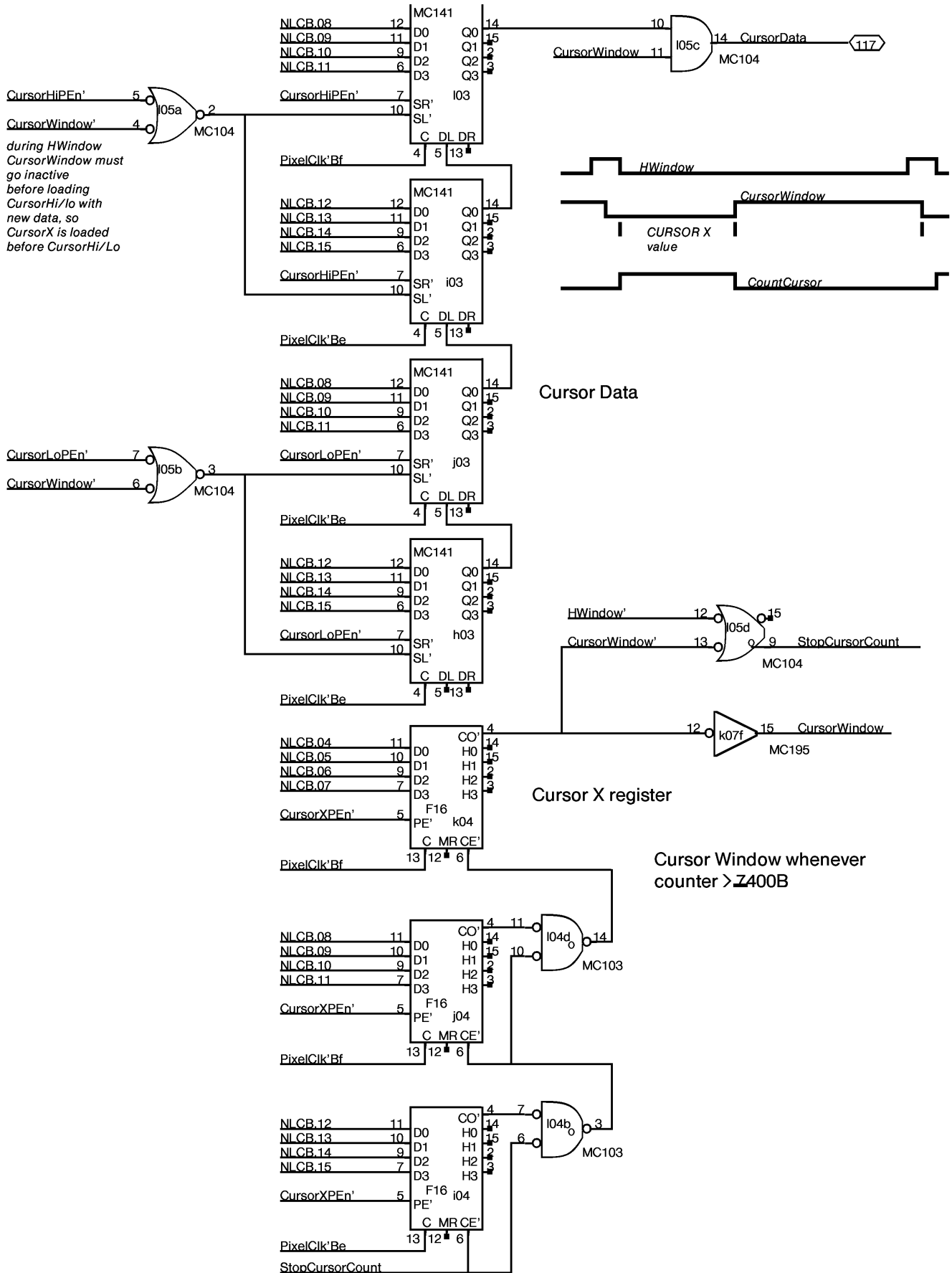
For DWT commands:

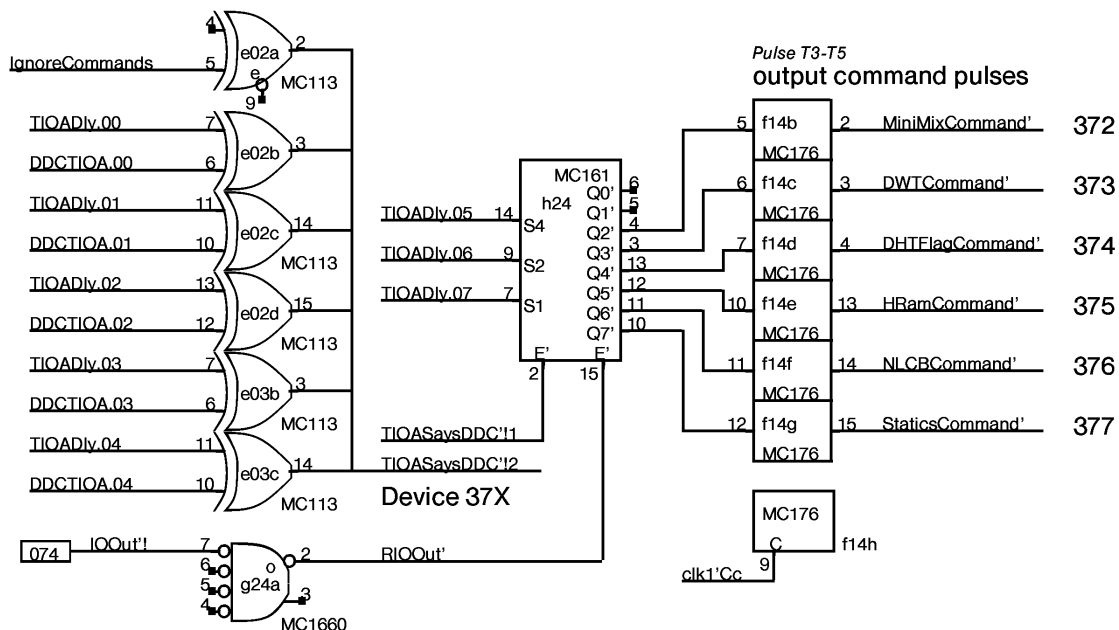
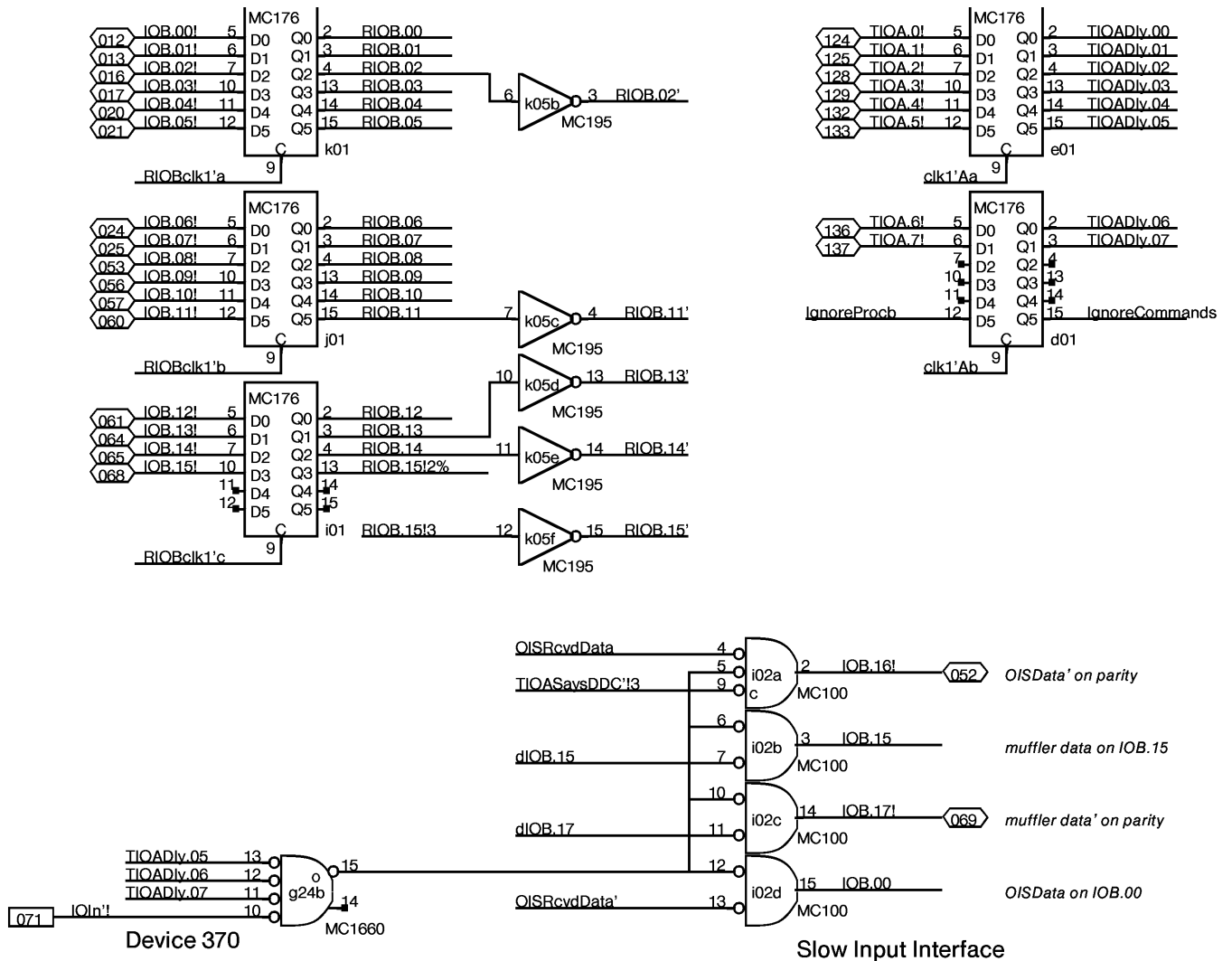
RI0B	Means
= 20c	IOFetch signal don't touch Flags
= 1c	Set CWCBFlag and Clear NWCBFlag
= 0c	Clear CWCBFlag

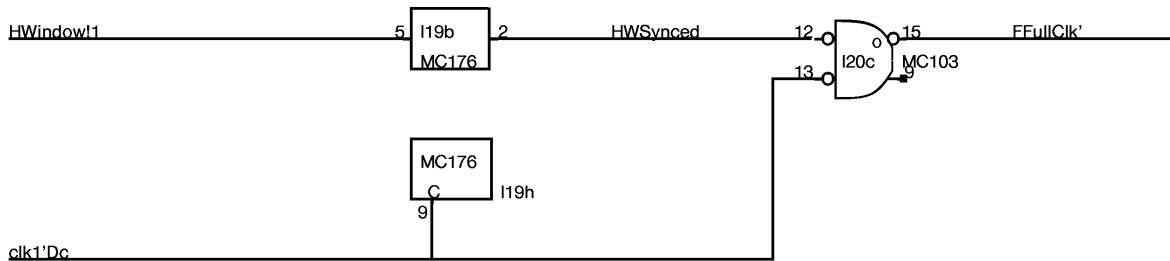
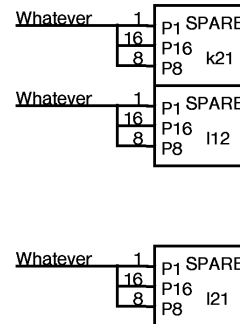
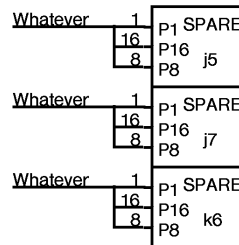
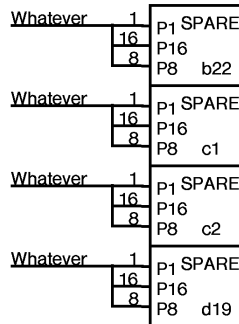
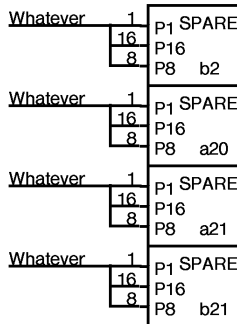
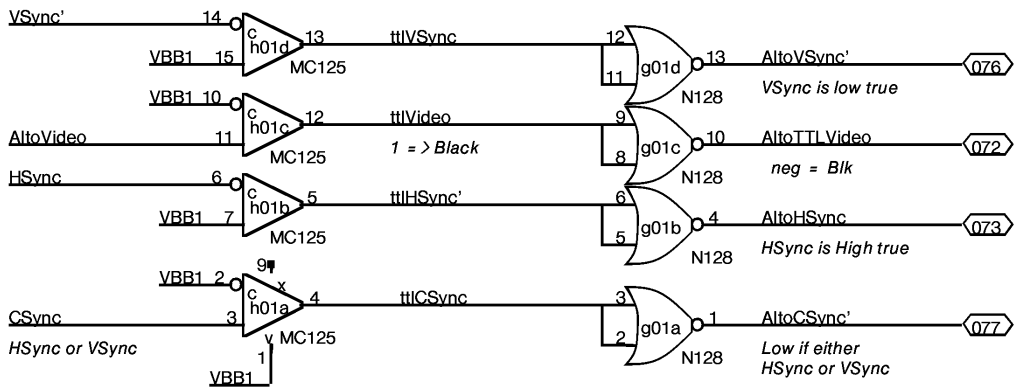
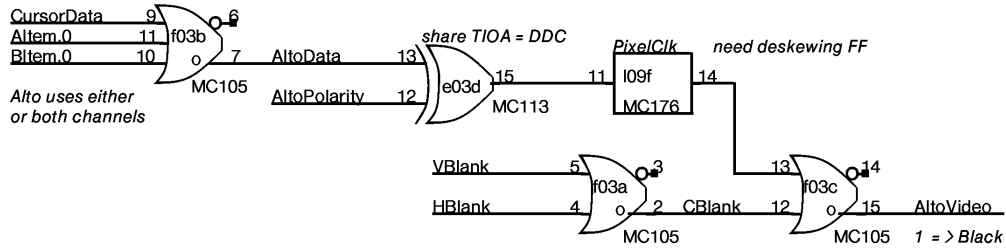
For DHT commands:

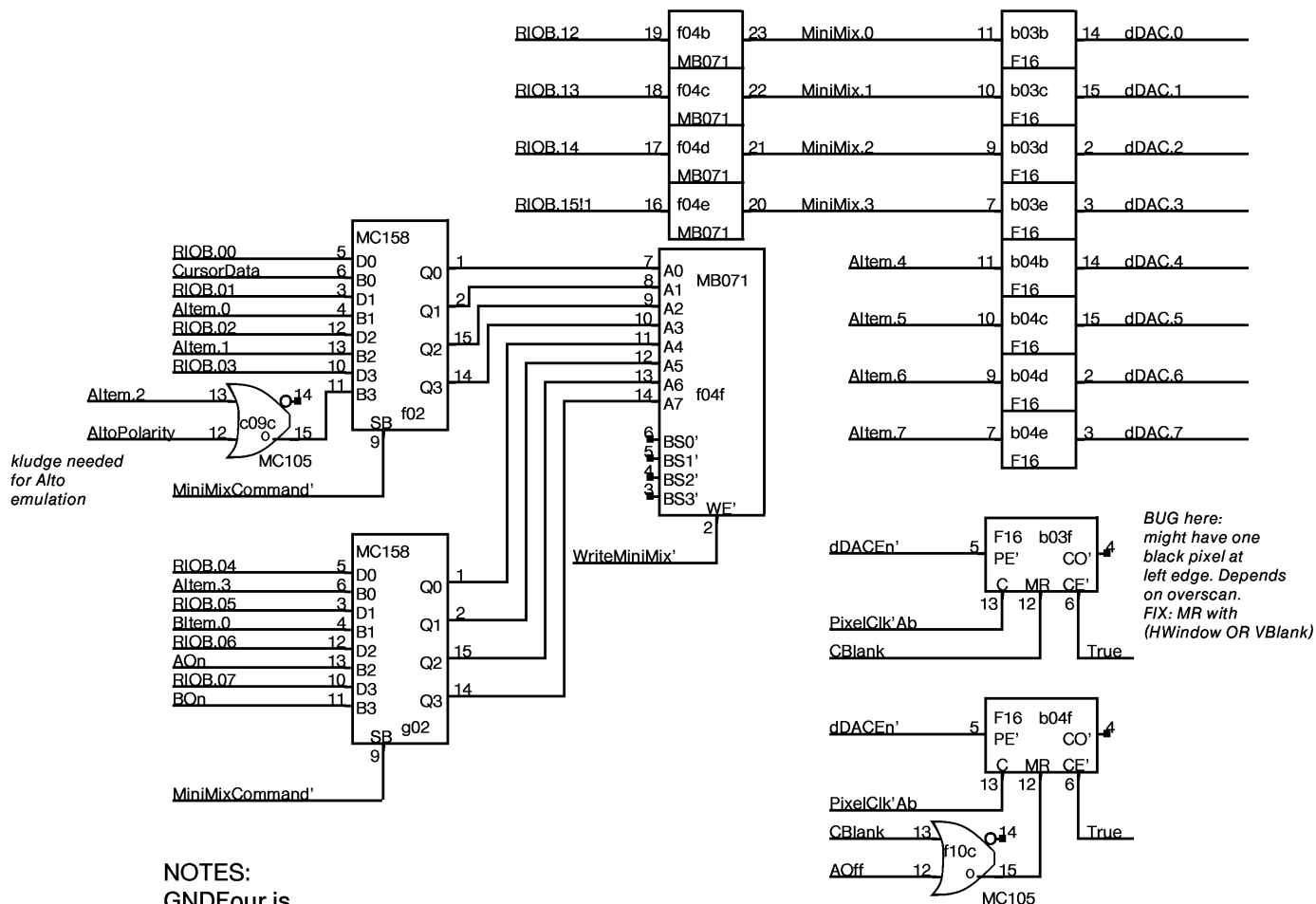
RI0B	Means
= 2c	Set ANextWCBFlag
= 4c	Set BNextWCBFlag



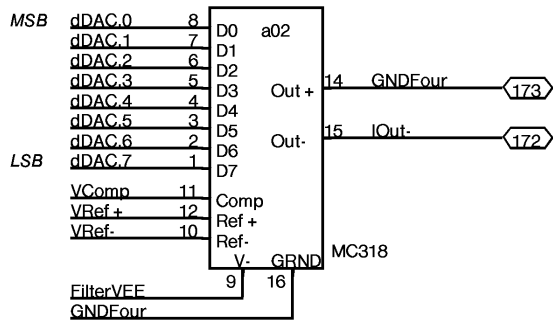




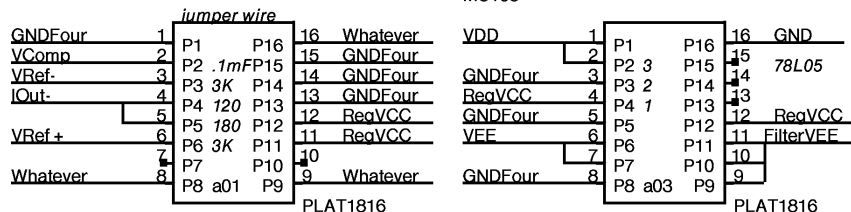




NOTES:
GNDFour is
used as single point GND
for DAC system



Digital/Analog converter



Platform is DAC components

1	jumper wire	16
2	0.1mFarad	15
3	3 KOhm	14
4	120 Ohm	13
5	180 Ohm	12
6	3 KOhm	11
7		10
8		09

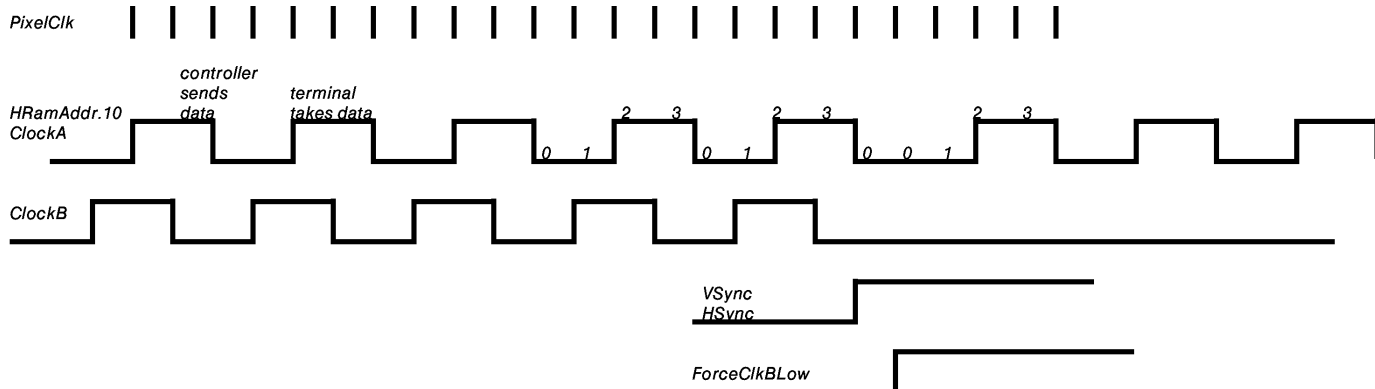
location a01

Voltage filters

1	0.1mFarad	16
2	in ³	15
3	78L05	14
4	out ¹	13
5	0.1mFarad	12
6	12 mHnry	11
7	2 ohm	10
8	+ 22 mF	09

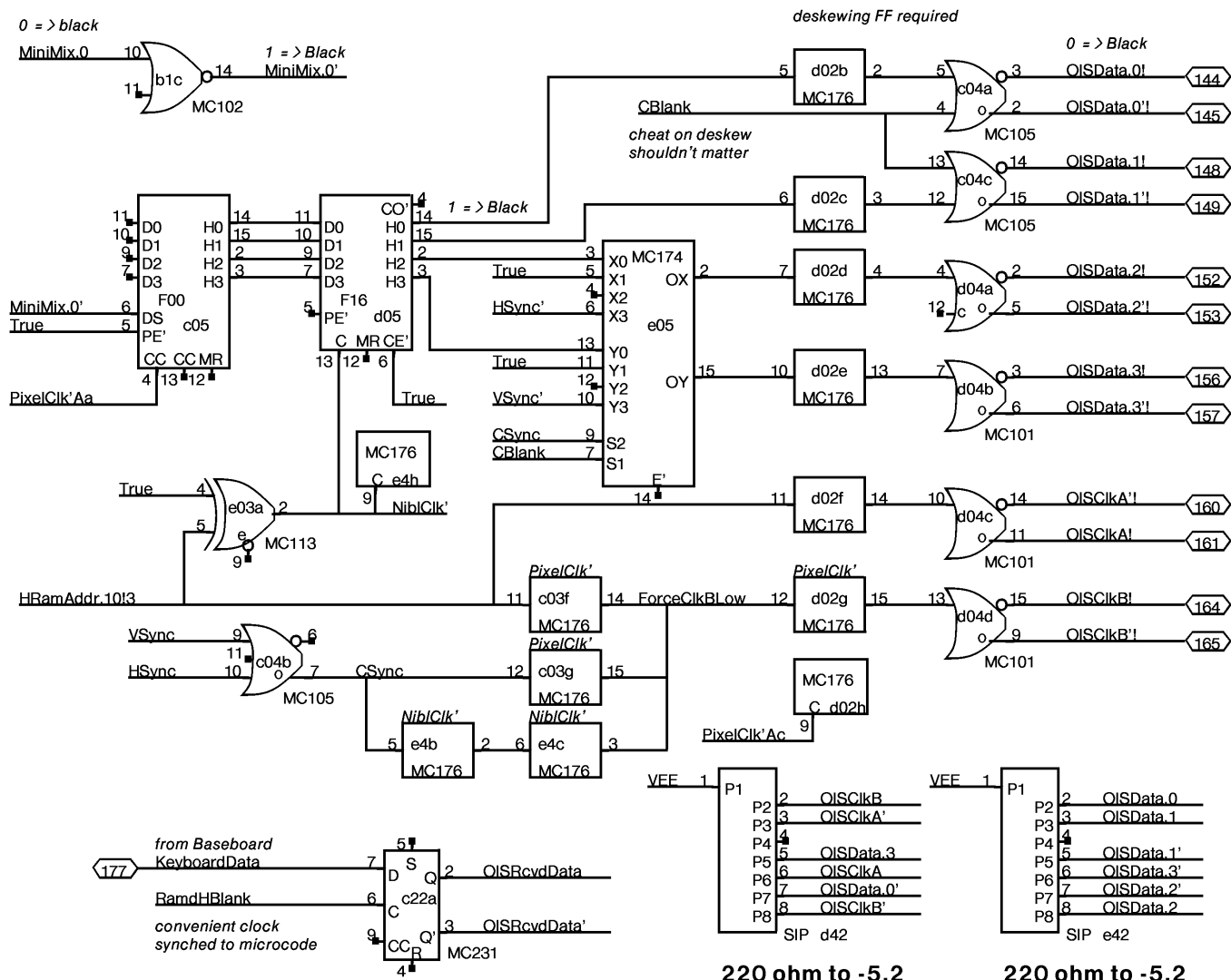
location a03

RevCd: swapped positions
of 12mH and 2mF



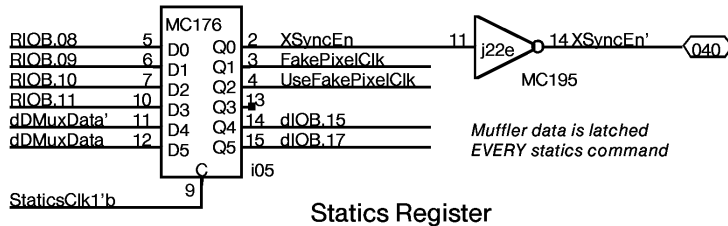
CLAIM: HRamAddr.10 is the waveform for ClockA. Delay it by one pixelclock, and invert, then you get the waveform for ClockB. ClockB is also jammed low by (HSynC OR VSynC) delayed by one PixelClk. To avoid hiccups in the phasing of ClockA with respect to ClockB, HSynC should only be generated just after HRamAddr.10-11 = 3, that is, a hiccup would occur if HRamAddr.01-11 = 1 just before HSynC occurred.

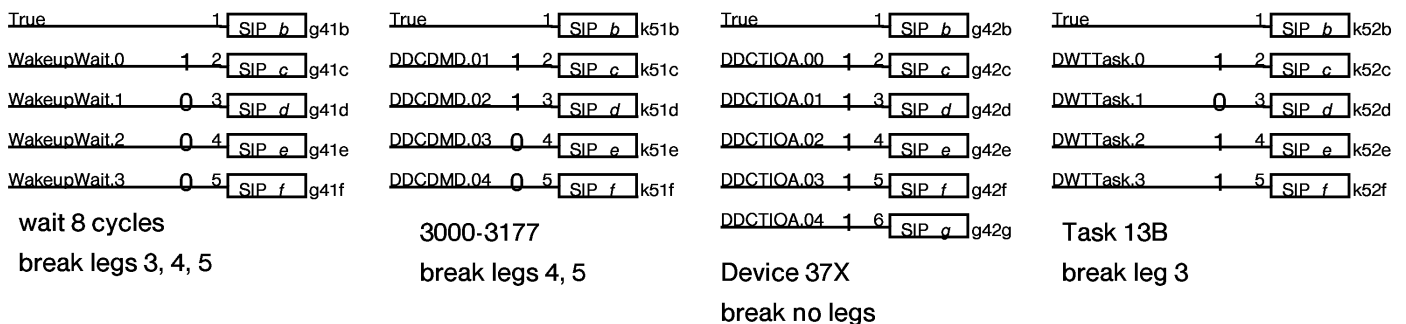
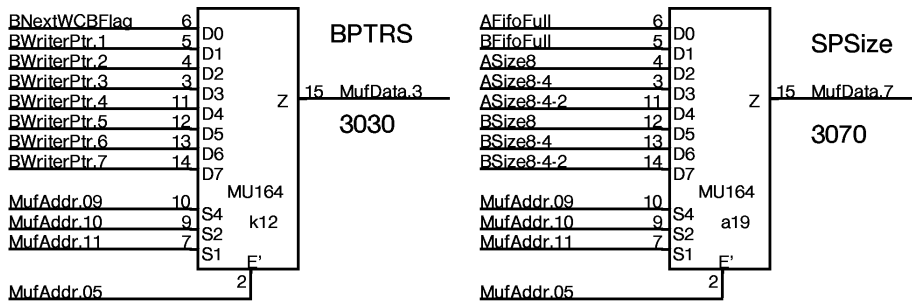
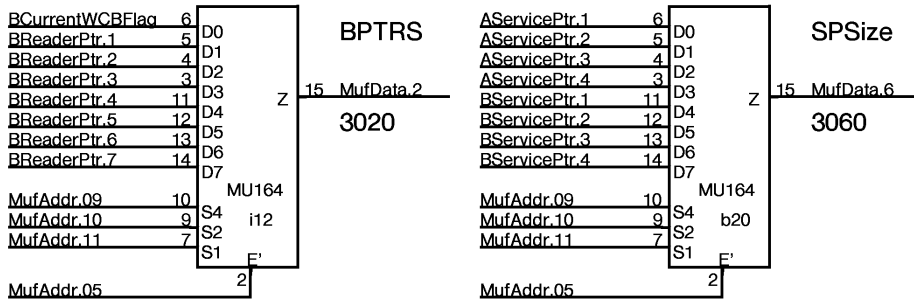
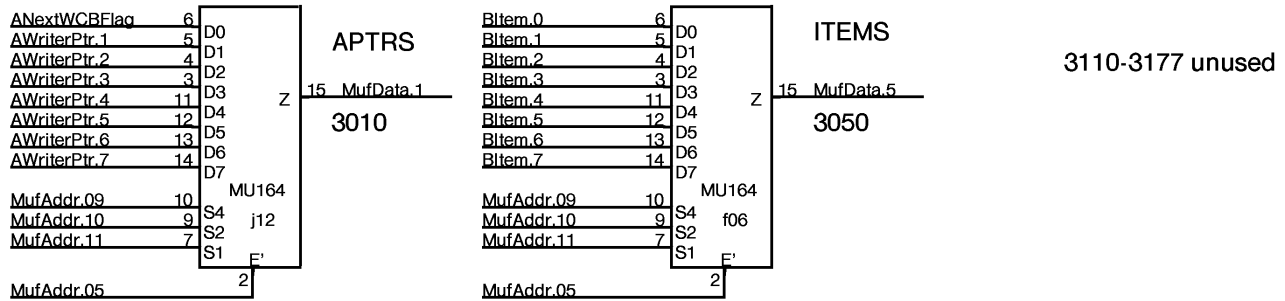
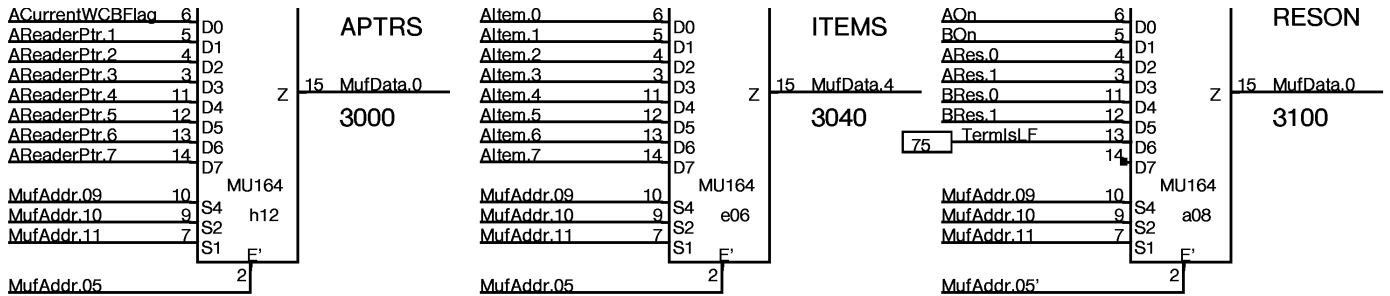
CBlank must be phased so that it changes when HRamAddr.10 goes from 1 to 0. However, CBlank is delayed by two pixel clocks from RamdHBlank, so the HRam must be programmed to make RamdHBlank change on the 0 to 1 transition of HRamAddr.10. This is slightly confusing, so be careful.



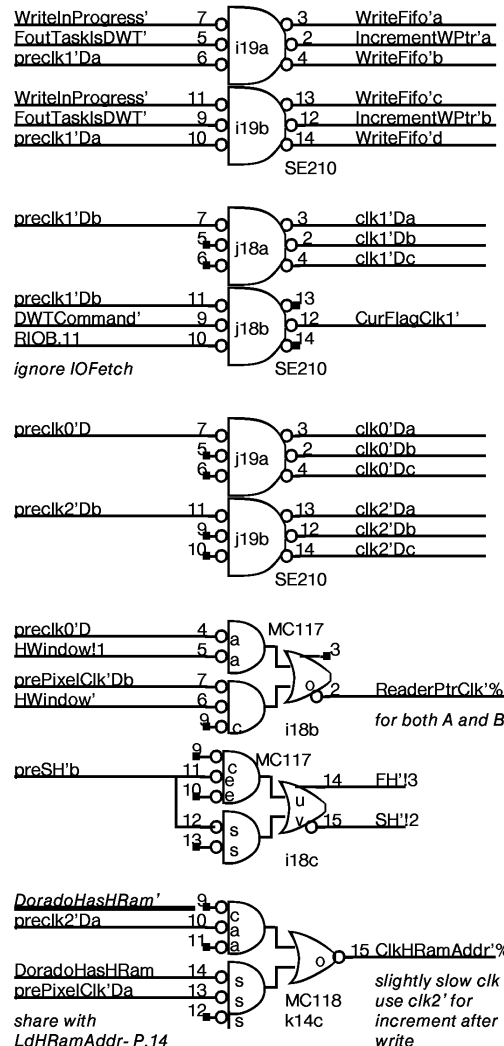
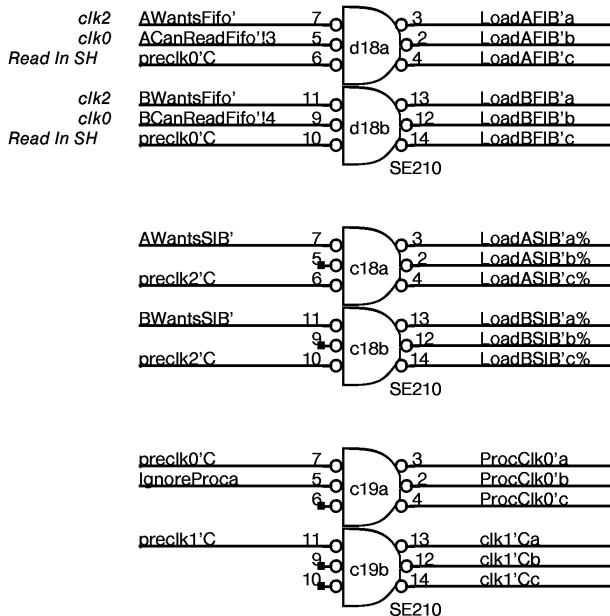
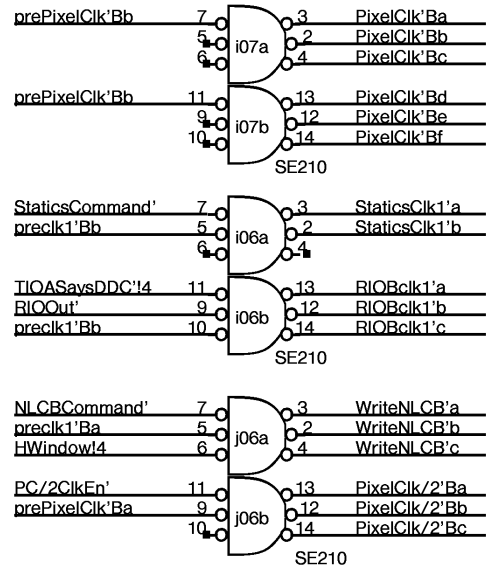
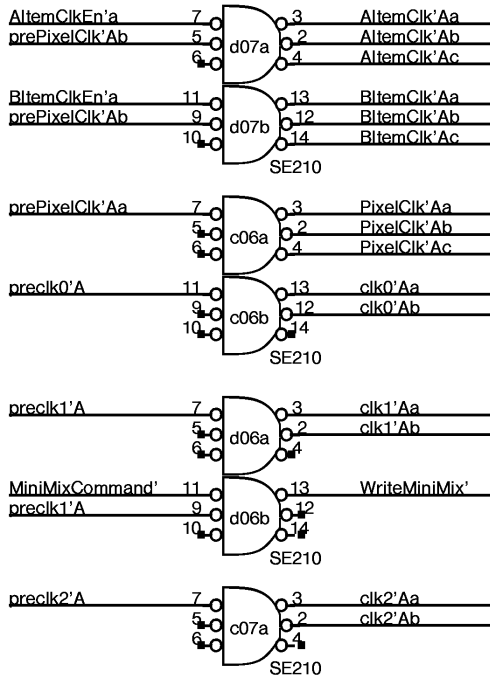
220 ohm to -5.2

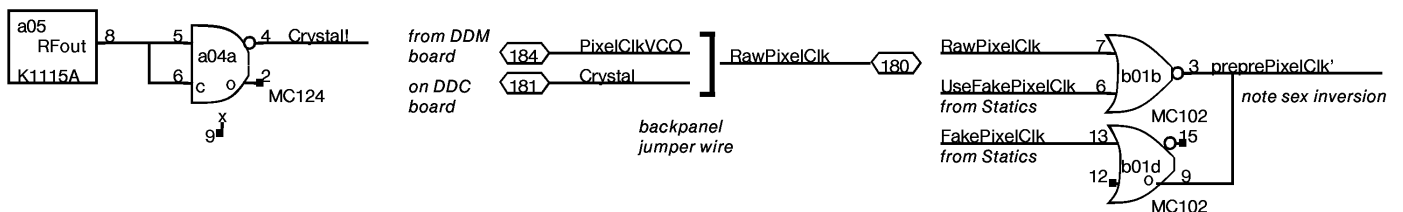
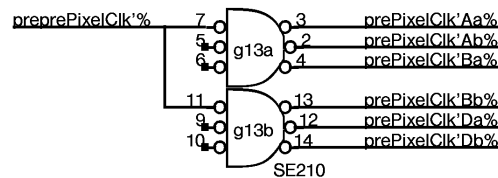
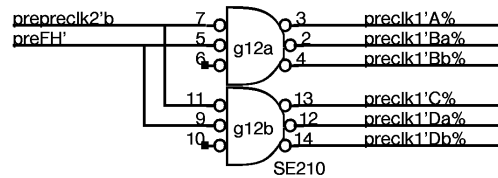
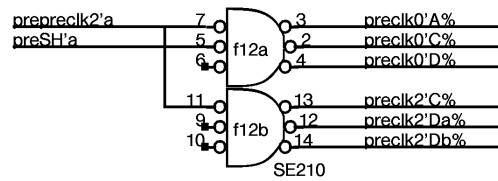
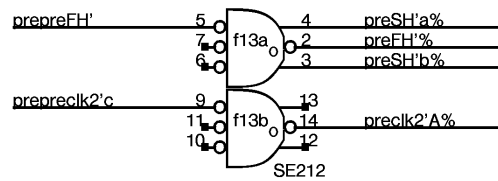
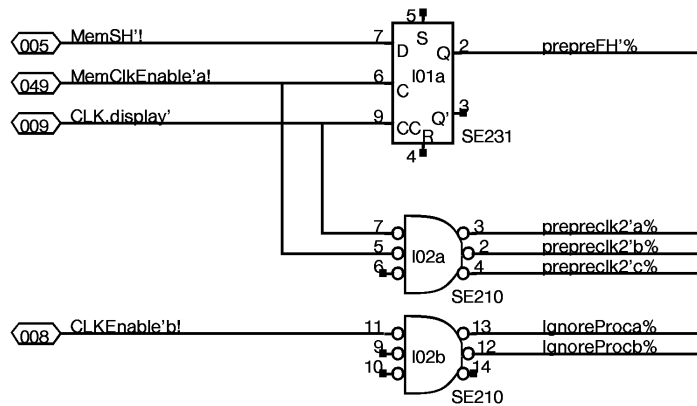
220 ohm to -5.2





All SIPs are standard 100 ohm terminators with legs broken as shown





CONNECTOR		CONNECTOR		CONNECTOR		CONNECTOR		CONNECTOR		CONNECTOR		CONNECTOR		CONNECTOR		CONNECTOR	
a 181		b 168		c 153		d 137		e 124		f 109		g 93		h 80		i 64	
1	DAC Plat	PCLK B>Ct 102			TIOADEF 176 1	TIOA 176 1	DHTTW RdFifo 135		ALTO 74128	ALTO 125	RIOB 176 rc1	RIOB 176 rc1	RIOB 176 rc1		SE231	1	
2	DAC 10318				OIS 176	TIOA = D 113	MMMux 158	MMMux 158	NLCB 145 W	DIQB 100	NLCB 145 W	NlcbAd F16 P2		SE210		2	
3	DAC Plat	MMix F16 PC	OIS ICEn 176 PC		DWTTw 231 0	TIOA = D 113	ALTO 105	NLCB 145 W	CURSR 141 P	CURSR 141 P	CURSR 141 P	NlcbAd 197		CURSR 141 P		3	
4	PCLK 124 BCt	MMix F16 PC	OIS 105		OIS 101	OIS 176	MMix	ClcbEn 161	VSyC 104	CURX F16 P	CURX F16 P	CURX F16 P		CURX 103		4	
5	PCLK Xtal	LSR 104	OIS F00		OIS F16	OIS 174	MMix	ClcbEn 161	STAT 231 sc1	STAT 176 sc1			195 RIOB'		CURSR 104	5	
6	AdICEN 121	BdICEn 121	PC MMix		0,1	MU AITEM	MU BITEM	ASIZE F16 P2	ARES F16 P2	sc1 rc1	NLCB 1 PC/2			MIDAS 158		6	
7	ARESC F16 P	BRESC F16 P	2 210 B		ABITEMCLK	AITEM 104	BITEM 104	BSIZE F16 P2	BRES F16 P2	PC PC			PULSE 195	MIDAS 158		7	
8	MU ABOFF OIS RES	LSR 118	ASRC F16 I		BSRC F16 I	AITEM 104	BITEM 104	HWNDQ 135 P2	BWDTH F16 P	BWDTH F16 P	BWDTH F16 P	BWDTH F16 P		MIDAS 164		8	
9	ASYNC 231 2	ASYNC 231 2	ASYNC 105		BSYNC 231 2	ASYNC 135 I	BSYNC 135 I	MODE F16 P2	ABOFF 135 P	AWDTH F16 P	AWDTH F16 P	AWDTH F16 P		176 PC P		9	
10	ASR I F00	BSR I F00	ASR F00		BSR I F00	ASR I F00	BSYNC 105	VCW F16 P2	BLMarg F16 P	BLMarg F16 P	BLMarg F16 P	BLMarg F16 P		HRmOut F16 P		10	
11	ASR I F00	BSR I F00	ASR F00		BSR I F00	ASR I F00	BSR I F00	WANT 117	ALMarg F16 P	ALMarg F16 P	ALMarg F16 P	ALMarg F16 P		AWDTH 103		11	
12	ASR I F00	BSR I F00	ASR F00		BSR I F00	ASR I F00	pc0' pc2' SE210	pc1' pc1' SE210	MU ARP AF	MU BRP BF	MU AWP AF	MU BWP BF				12	
13	ASIB 176	BSIB 176	AFIB F00		BFIB F00	BFIB F00	pc2 FHSB SE212	ppc' ppc' SE210	HRAddr F16 H	HRAddr F16 H	HRAddr F16 H	HRAddr F16 H		MIDAS 166		13	
14	ASIB 176	BSIB 176	AFIB F00		BFIB F00	BFIB F00	COMM' 176 1	flags 117	HRam 415 HW	HRam 415 HW	HRam 415 HW	HRamC 118		HRamC 118		14	
15	ASIB 176	BSIB 176	AFIB F00		BFIB F00	IP	NFlags 231 1	CFlags 231 cf	DHasHR 231 1,p2	AFULL 149	BFULL 176 1	AFULL 176 1		BFULL 149		15	
16	ASIB 176	BSIB 176	AFIB F00		BFIB F00	IP	IP			ARP rpc F16	BRP rpc F16	AWP wpc F16		BWP wpc F16		16	
17	ASIB 176	BSIB 176	AFIB F00		BFIB F00	IP	IP	FIFO	FIFO	ARP rpc F16	BRP rpc F16	AWP wpc F16		BWP wpc F16		17	
18	ASIB 176	BSIB 176	ABSIB 2 ABSIB 2		ABFIB 0	IP	IP			FHSB rpc 0 117	1, cflg 1	FAM 174		PTRS 103		18	
19	MU BA-SZE	ABSize 103	PROCO' 1'			IP	IP	FIFO	FIFO	fifo 1 wpc 1	2 0	FAM 174		Sync 176		19	
20		MU SP	AFIB F00		BFIB F00	IP	IP			FA 2 231	FAM FH 174 SH	FAM 174		IncSP 103 c		20	
21			AFIB F00		BFIB F00	IP	IP	FIFO	FIFO	FA 2 231	FA 2 231					21	
22	CTDWT 117 c		OIS 231		IP	IP	IP			FA 2 231	d 195	MIDAS 176		MIDAS 176		22	
23	102 c	clk0 176 FG	GETS 231 1		BLKDB 135 1	ASP 1 F16	BSP 1 F16	FIFO	FIFO	FOUT 1 176	FOUT 1 176	FOUT 1 176		MIDAS 102 BC		23	
24	NEXT 113	SUBT 106 A	105		195	WSpace F16 0	HELD 231 0	IOInOut 1660	TIOA = D 161	FOUT 0 176	FOUT 0 176	FOUT 0 176		FTsk 113		24	
CONNECTOR		CONNECTOR		CONNECTOR		CONNECTOR		CONNECTOR		CONNECTOR		CONNECTOR		CONNECTOR		CONNECTOR	
a 11		b 26		c 39		d 55		e 70		f 86		g 99		h 114		i 129	
NEXT		SubT BLK		FIN		Hold IOIn/out		CONNECTOR		IOIn FNXT		FOUT		FTsk		DMux	
XEROX PARC		Project Dorado		Reference DDC Board Layout				File DispY26.sil		Designer K. Pier		Rev Cj		Date 3/26/81		Page 26	

DDC Slow IO System

DEVICE	TIOA	I/O	TASK	FORMAT and COMMENTS
STATICS	377	O	DHT, EMU	See Below
NLCB	376	O	DHT, EMU	NLCBAddr.0-3,,Data.4-15
HRAM	375	O	DHT, EMU	Keep,Write,LoadAddr,0,Data.4-15
DHTFLAG	374	O	DHT	ANextWCBFlagSet,BNextWCBFlagSet,0 bits 13,14,15
DWT	373	O	DWT	IOFetch signal,.....,Set/Clr CWCBFlagANDClrNWCBFlag bit 11, , bit 15
MiniMixer	372	O	DHT, EMU	address,,data always writes MiniMixer RAM
STATUS	370	I	DHT, EMU	Selected muffler input returned in bit 15
PIXELCLK	367	O	DHT, EMU	Pixel clock rate located on mixer board
MIXER	366	O	DHT, EMU	Keep,Write,LoadAddr,0,Data.4-15 located on mixer board

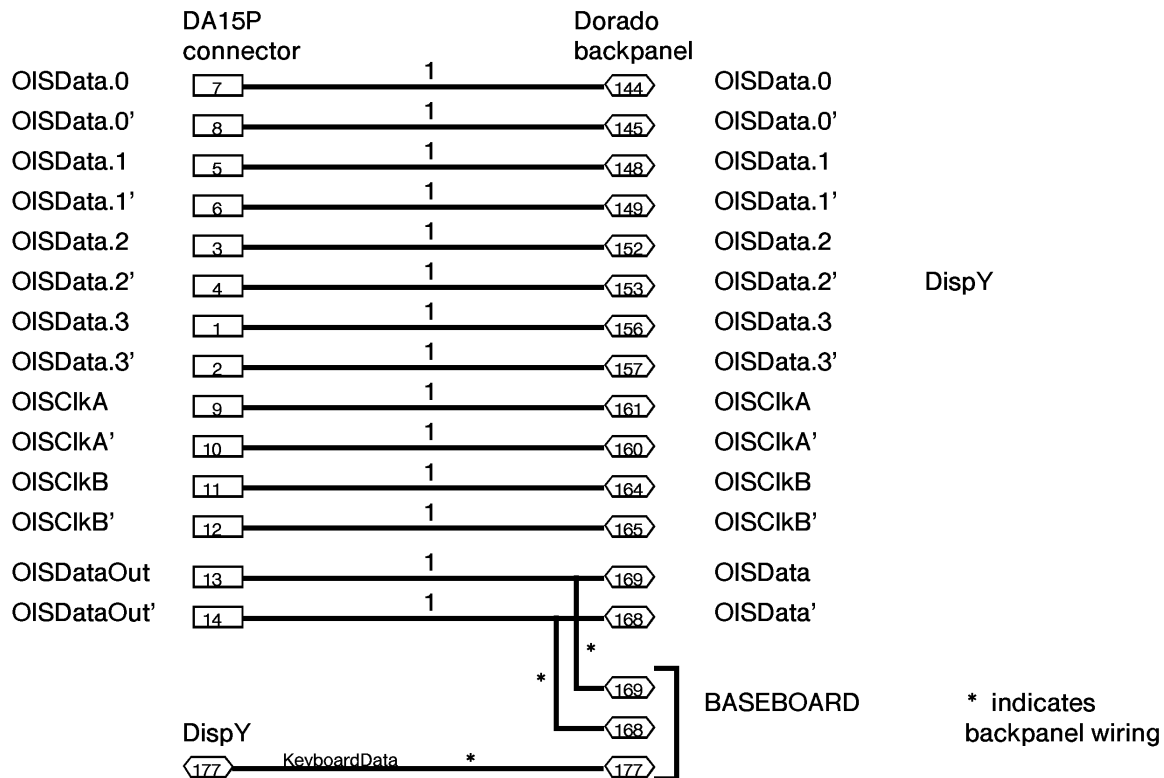
00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	
X	Muff Addr .05	Muff Addr .06	Muff Addr .07	Muff Addr .08	Muff Addr .09	Muff Addr .10	Muff Addr .11	XSync Enable	Fake Pixel Clock	Use Fake Pixel Clock	Unused	Unused	Unused	DWT Shut Up	DHT Shut Up- Reset DDC	Statics
NLCB Addr .00	NLCB Addr .01	NLCB Addr .02	NLCB Addr .03	NLCB Data 12 bits												NLCB
Keep HRam'	Write HRam'	Load HRam Addr				Addr.0	Addr.1	Addr.2	Addr.3	Addr.4	Addr.5	Addr.6	Addr.7	Addr.8	Addr.9	HRAM
													HSync	HBlank	Half Line	
													Set BNext WCB Flag	Set ANext WCB Flag	Must Be 0	DHTFLAG
											IOFetch signal	Must Be 0	Must Be 0	Must Be 0	Set/ Clr Cur WCB Flag	DWT
Address.0-7								Data.0-7								MiniMixer
								ClkRate.0-7								PIXELCLK
Keep Mixer'	Write Mixer'	Load Mixer Addr	X		Addr.0	Addr.1	Addr.2	Addr.3	Addr.4	Addr.5	Addr.6	Addr.7	Addr.8	Addr.9	Hi/Lo select	MIXER
												Mixer Data 12 bits				

Next Line Control Block Format

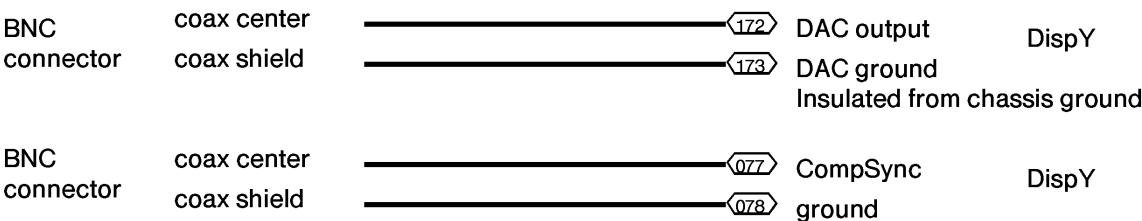
Address	Name	Format
0	VCW vertical control word	0...0, VBlank, VSync, EvenField
1	AMargin	LMarg[00..11]
2	AWidth	Width[00..11]
3	AFifoAddr	FifoAddr[0..7] *must be even
4	AScan	0...0,Polarity,Resolution[0..1],Size8,Size4,Size2,Size1
5	ModeControl	0..0,24Bit,ABypass,BBypass,A8B2
6	reserved	
7	reserved	
10	unused	
11	BMargin	LMarg[00..11]
12	BWidth	Width[00..11]
13	BFifoAddr	FifoAddr[0..7] *must be even
14	BScan	0...0,Polarity,Resolution[0..1],Size8,Size4,Size2,Size1
15	CursorX	CursorXCount[0..11]
16	CursorLo	CursorLoByte[4..11]
17	CursorHi	CursorHiByte[4..11]

DDC	SIZE	DDM
Altem	8	Altem 85-88-89-92-93-96-97-100
Bltem	8	Bltem 101-104-105-108-109-112-113-116
CursorData	1	CursorData
AltemClkEn	1	AltemClkEn
BltemClkEn	1	BltemClkEn
AOff	1	AOff
BOff	1	BOff
HSync	1	HSync
HBlank	1	HBlank
HalfLine	1	HalfLine
VSynC	1	VSynC
VBlank	1	VBlank
PixelClk	1	PixelClk
Crystal	1	Crystal
Oscillator	1	Oscillator
Modes	4	Modes 28-29-32-33
XHSync	1	XHSync
XVSync	1	XVSync
XSyncEn	1	XSyncEn
OISData.0	1	7 wire interface cable
OISData.0'	1	
OISData.1	1	
OISData.1'	1	
OISData.2	1	
OISData.2'	1	
OISData.3	1	
OISData.3'	1	
OISClkA	1	
OISClkA'	1	
OISClkB	1	
OISClkB'	1	
OISDataOut	1	
OISDataOut'	1	
AltoVideo	1	Alto monitor cable
AltoHSync	1	
AltoVSync	1	
AltoCSync	1	Grey level driver cables 2 coax
DAC	1	
DAC GND	1	
spare	1	from Baseboard
KeyboardData	1	

Seven Wire Interface Cable



Grey level cabling



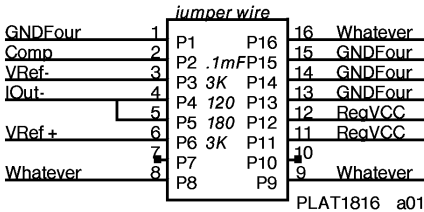
Note: CompSync currently appears in the Alto cable. The Alto display doesn't make use of CompSync. We could simply replace the CompSync pair in the Alto cable with the coax. We then have the capability to attach a local Alto type display and a remote display.

Six BNC connectors mounted on the "square" side. Cabling not yet defined.

1. Plug an MC10318 D/A converter into location a02 .
2. Plat1816 in locations a01 and b02 are discrete components, shown below.
3. SIPs are 100 ohm terminator package with legs broken:

location	break legs	See page 23.
g41	3,4,5	
g42	none	
k51	4,5	
k52	3	

4. SIPs at locations d42 and e42 are 220 ohm value instead of 100 ohm.
5. Crystal oscillator K1115A, location a05, value 20 MHz.

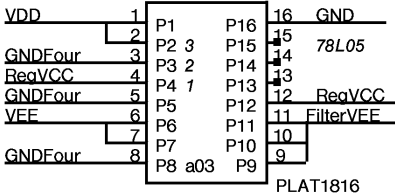


Platform is DAC components

1	jumper wire	16
2	0.1mFarad	15
3	3 KOhm	14
4	120 Ohm	13
5	180 Ohm	12
6	3 KOhm	11
7		10
8		09

location a01

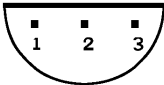
See Page 20.



Voltage filters

1	0.1mFarad	16
2	in ³	15
3	gnd ²	14
4	out ¹	13
5	0.1mFarad	12
6	12 mHnry	11
7	2 ohm	10
8	+ 22 mF	09

location a03



Bottom view of pinout for 78L05